

TECHNICAL HANDBOOK FOR INFORMATION, USE AND MAINTENANCE

Machine nr.	: A642
Customer	: AUTOMOTIVE LIGHTING
Factory	: INDIA
Description	: HOT PLATE WELDING MACHINE 3AX

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1. SAFETY

The regulations as per this handbook refer to operations allowed on machine: instructions must be followed to avoid operator's injury and possible damages to the machine.

1.1 QUALIFIED PERSONNEL

Operators and maintenance people are considered to be QUALIFIED PERSONNEL when they have been trained to achieve the necessary experience to use, maintain and repair the machine in whole safety, in respect of regulation.

1.2 SAFETY SYMBOLS



WARNING

IF NOT CONSIDERED SERIOUS DAMAGES UP TO DEATH MAY OCCUR TO PERSONS OR TOOLS.



WARNING

DANGER OF ELECTRIC SHOCK. IF NOT CONSIDERED, SERIOUS DAMAGE UP TO DEATH MAY OCCUR TO PERSONS OR TOOLS.



WARNING

HOT SURFACE. DANGER OF BURN. IF NOT CONSIDEREDS HEAVY BURNS MIGHT OCCUR

1.3 EC DECLARATION OF CONFORMITY

DECLARATION OF CONFORMITY CE N.: **003/2010**

THE MANUFACTURER: F.I. AUTOMAZIONE S.r.l.
Via Sommariva 2
12030 Caramagna Piemonte (CN).
Tel. ++39 (0172) 89561
Fax ++39 (0172) 89096
info@fiautomazione.com

DECLARES THAT THE UNDERMENTIONED NEW MACHINE:

Customer: AUTOMOTIVE LIGHTING
Reference nr.: A642
Description: Hot plate welding machine 3AX
Year of production: 2010

MEETS THE RELEVANT BASIC SAFETY AND HEALTH REQUIREMENTS OF EC DIRECTIVES:

89/392/CEE, 14 JUNE 1989, and following versions.
89/336/CEE, 3 MAY 1989, and following versions.

SIGNED BY : Gallo Silvio
POSITION : Managing Director

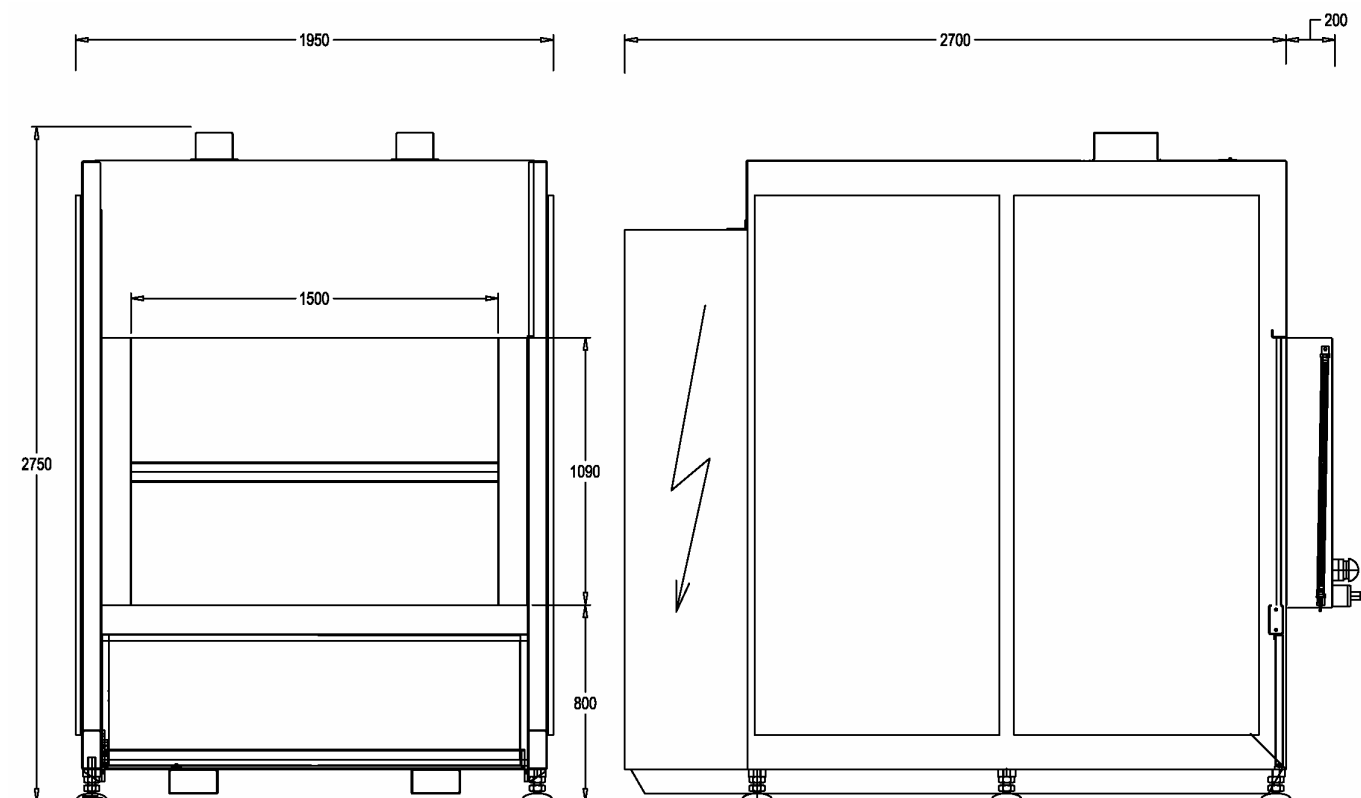
Caramagna Piemonte, 05/05/2010 Signature

2. MACHINE DESCRIPTION

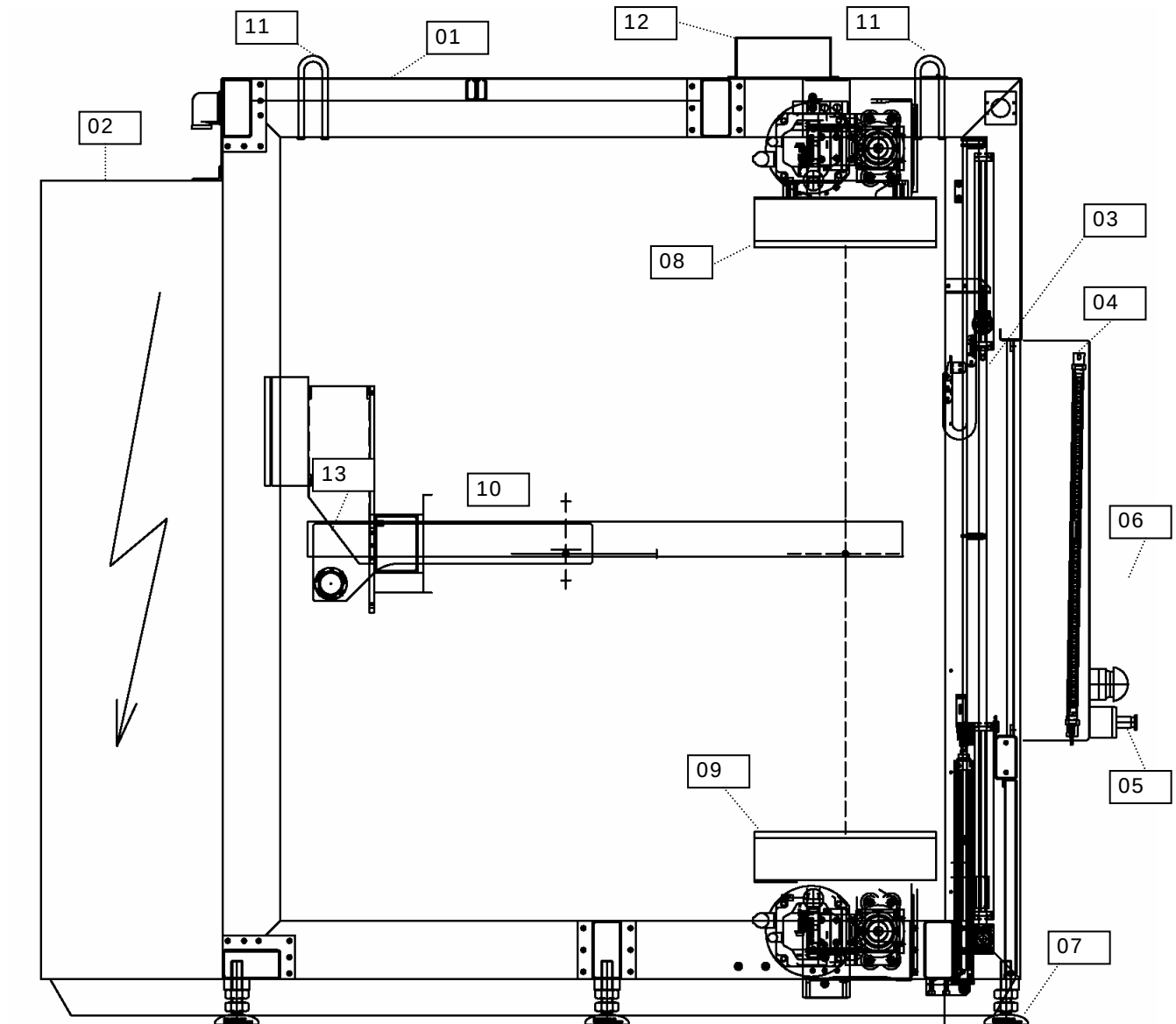
2.1 TECHNICAL DATA

SUPPLIER:	F.I. AUTOMAZIONE S.r.l. Via Sommariva 2 12030 Caramagna Piemonte (CN). Tel. ++39 (172) 89561 Fax ++39 (172) 89096
REFERENCE CODE:	A642
DESCRIPTION:	Hot plate welding machine 3AX
YEAR OF PRODUCTION :	2010
POWERING :	400 Vac 3PH + N + Earth
MAX ABSORPTION :	125 A
PNEUMATIC SUPPLY :	0.6 MPa (6 Bar)
MASCHINE WEIGHT :	3.500 Kg

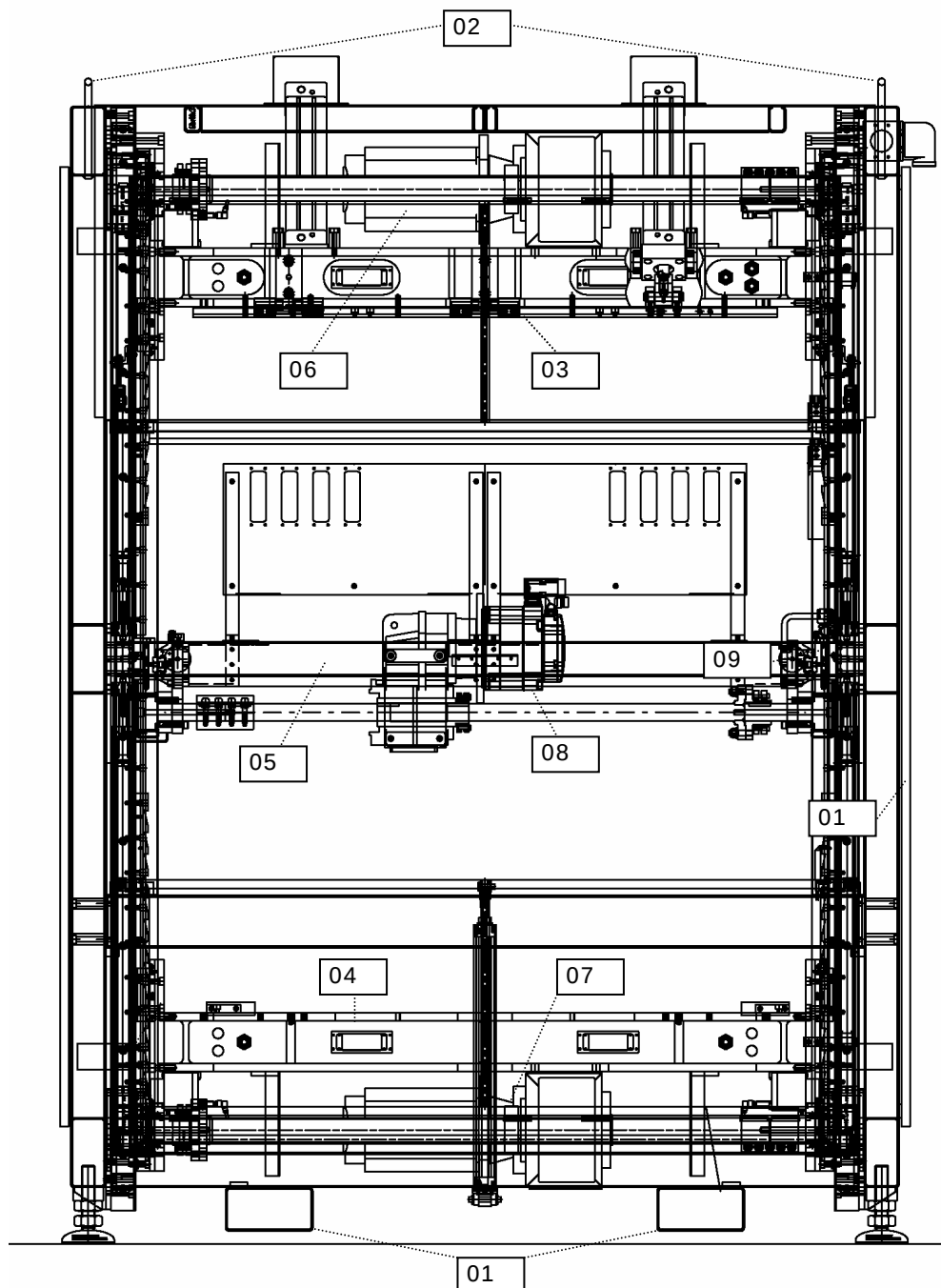
2.2 MACHINE DIMENSIONS



2.3 DESCRIPTION MACHINE PARTS



- 01 Structure in electrowelded steel profiles
- 02 Main electrical cabinet
- 03 Gate
- 04 Lightbarrier
- 05 Cycle Start button and Emergency button
- 06 Terminal operator and control keypad
- 07 Foot to set the height (nr.6)
- 08 Upper plate Axe AX1
- 09 Lower plate Axe AX2
- 10 Hot mirrors plate AXE AX3
- 11 Hooking for lifting the machine with crane
- 12 Connection gas suction



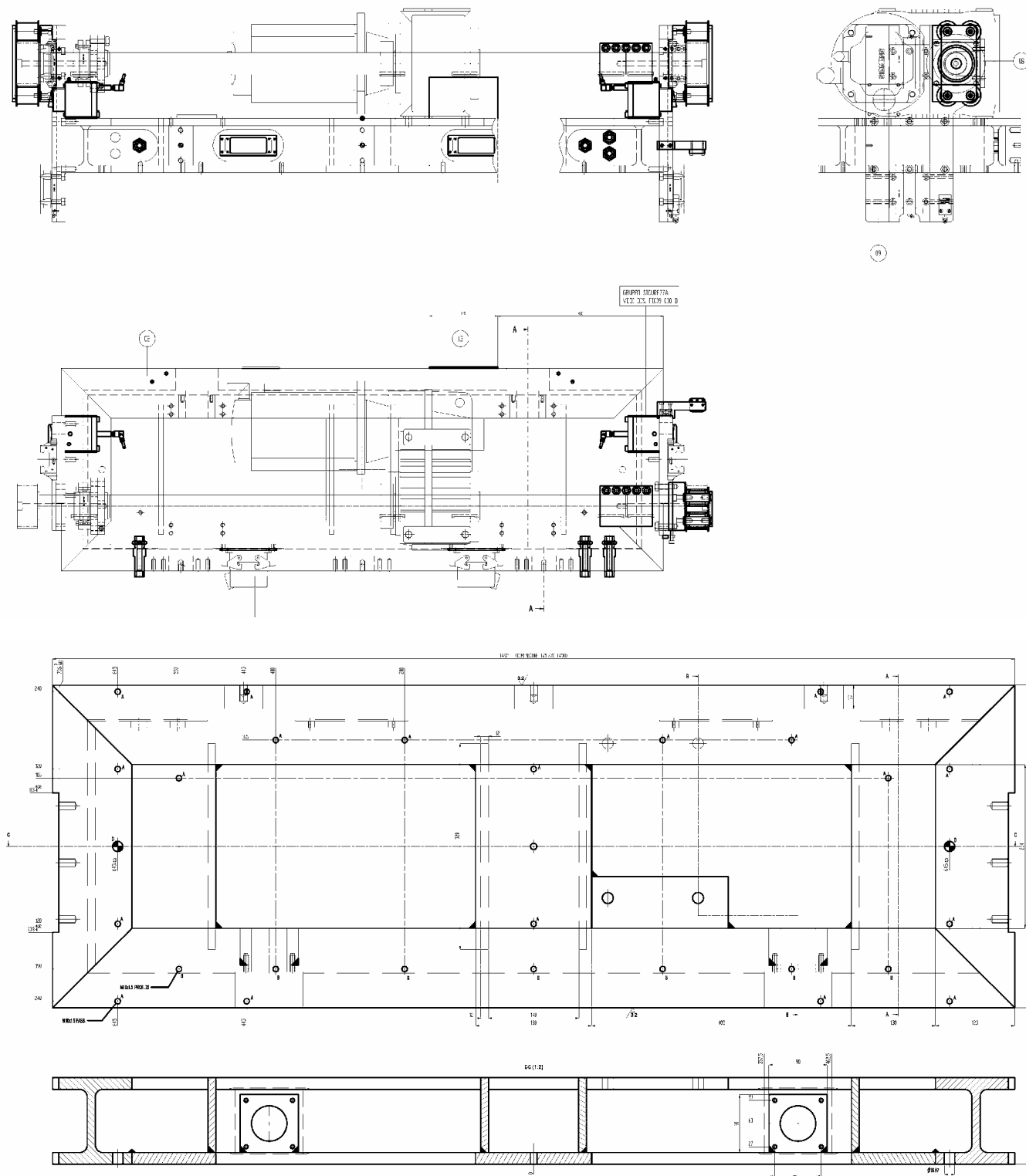
- 01 Fork holes for lifting the machine with forklift
- 02 Hooking for lifting the machine with crane
- 03 Upper plate Axe AX1
- 04 Lower plate AXE AX2
- 05 Hot mirrors plate AXE AX3
- 06 Upper plate motor AXE AX1
- 07 Lower plate motor AXE AX2
- 08 Motor hot mirrors plate AXE AX3

2.3.1 UPPER PLATE – AX1

The movement of vertical axes is given by a system that consists in Brushless engine / reducer / pinion / rack.
The max. stroke of vertical plates is 620 mm.
The max. load on the vertical plates is 200Kg.

On vertical plate there are two safety systems to avoid falling:

- STANDING BRAKE INTEGRATED IN THE ENGINE
- PNEUMATIC SAFETY KEY WITH RECOVERY

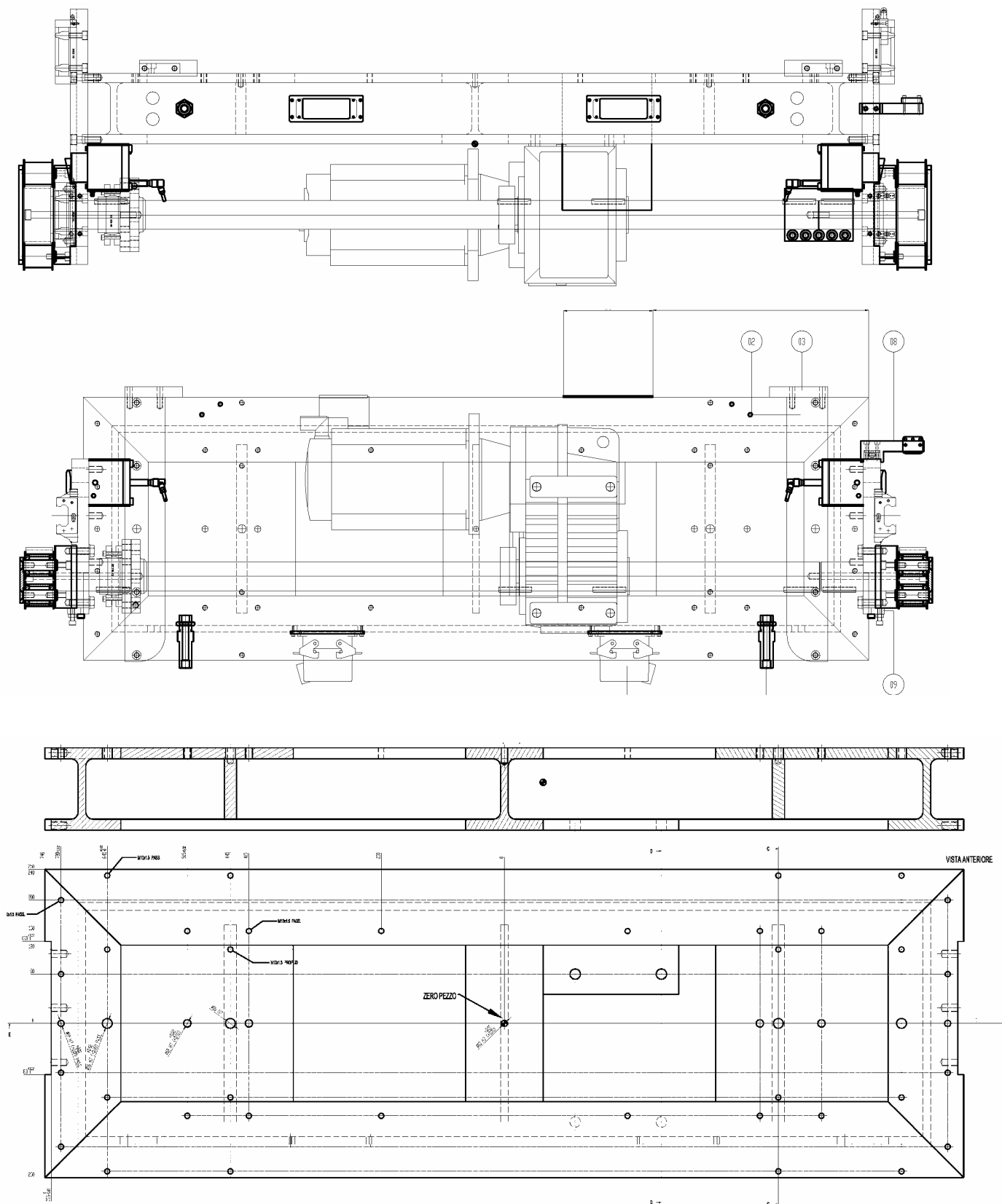


2.3.2 LOWER PLATE – AX2

The movement of vertical axes is given by a system that consists in Brushless engine / reducer / pinion / rack.
The max. stroke of vertical plates is 620 mm.
The max. load on the vertical plates is 200Kg.

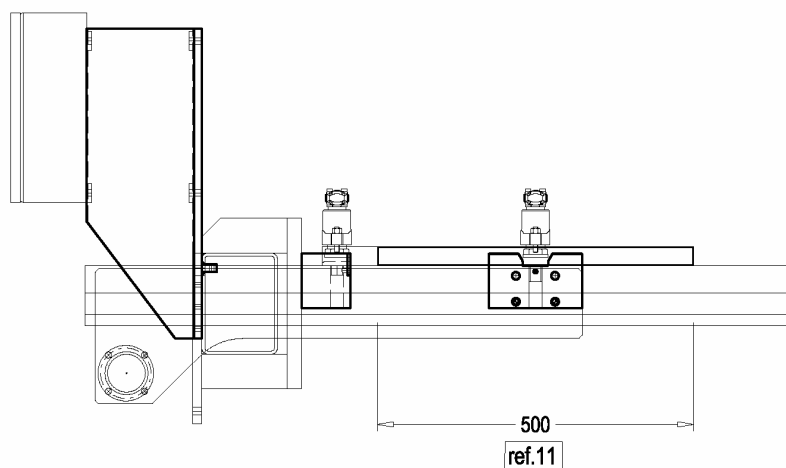
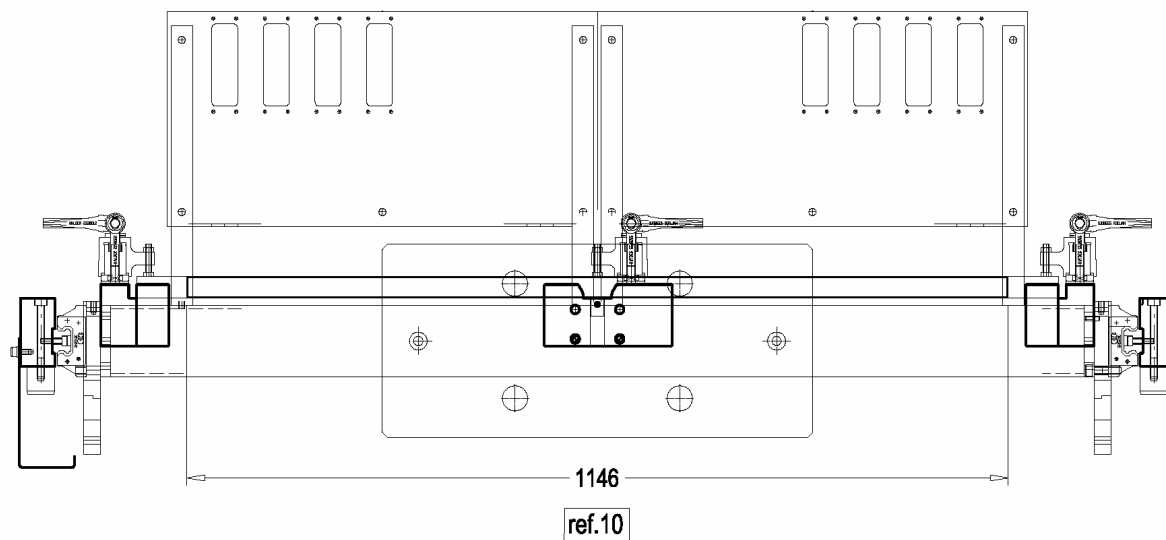
On vertical plate there are two safety systems to avoid falling:

- STANDING BRAKE INTEGRATED IN THE ENGINE
- PNEUMATIC SAFETY KEY WITH RECOVERY



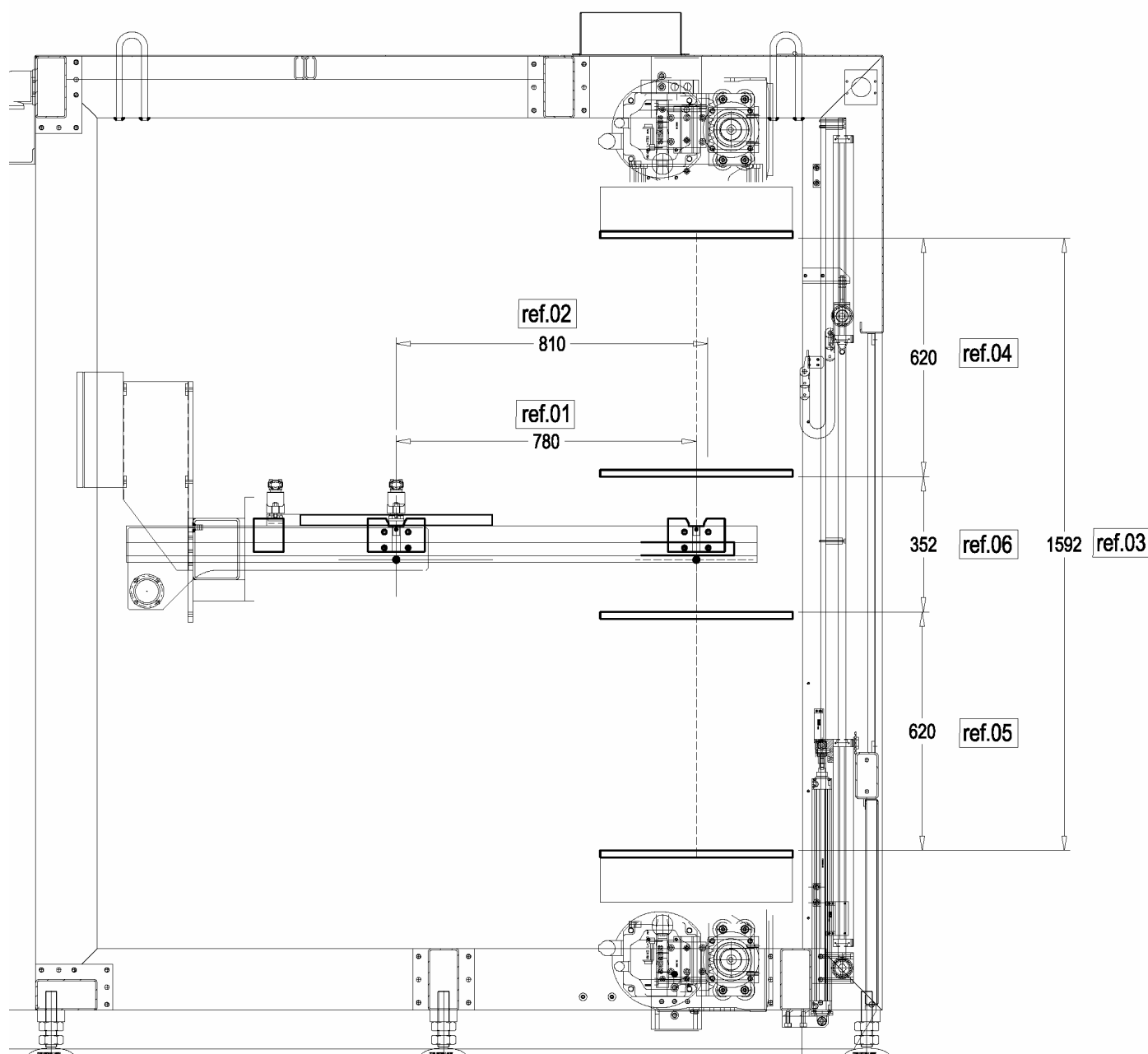
2.3.3 HOT MIRROR CHARIOT – AX3

The movement of the chariot is given by a system consisting in a Brushless ENGINE / reducer / pinion / rack.
The max. stroke of vertical plates is 810 mm.
The max. load on the vertical plates is 250Kg.



2.3.4 LIMITS OF AXES MOVEMENTS

- Ref.01 Theoretical stroke of hot mirror chariot
When the chariot is in the position 780,00mm, the centre line of mirrors and the center line of lower/upper fixture are aligned.
- Ref.02 Max. position of hot mirror chariot
- Ref.03 Max opening of the machine, measured between the upper/lower plates
- Ref.04 Max. stroke of the upper axe (the overstroke sensor at work intervenes at about 624mm)
- Ref.05 Max. stroke of the lower axe (the overstroke sensor at work intervenes at about 624mm)
- Ref.06 Min. closing of the machine measured between the upper/lower fixtures

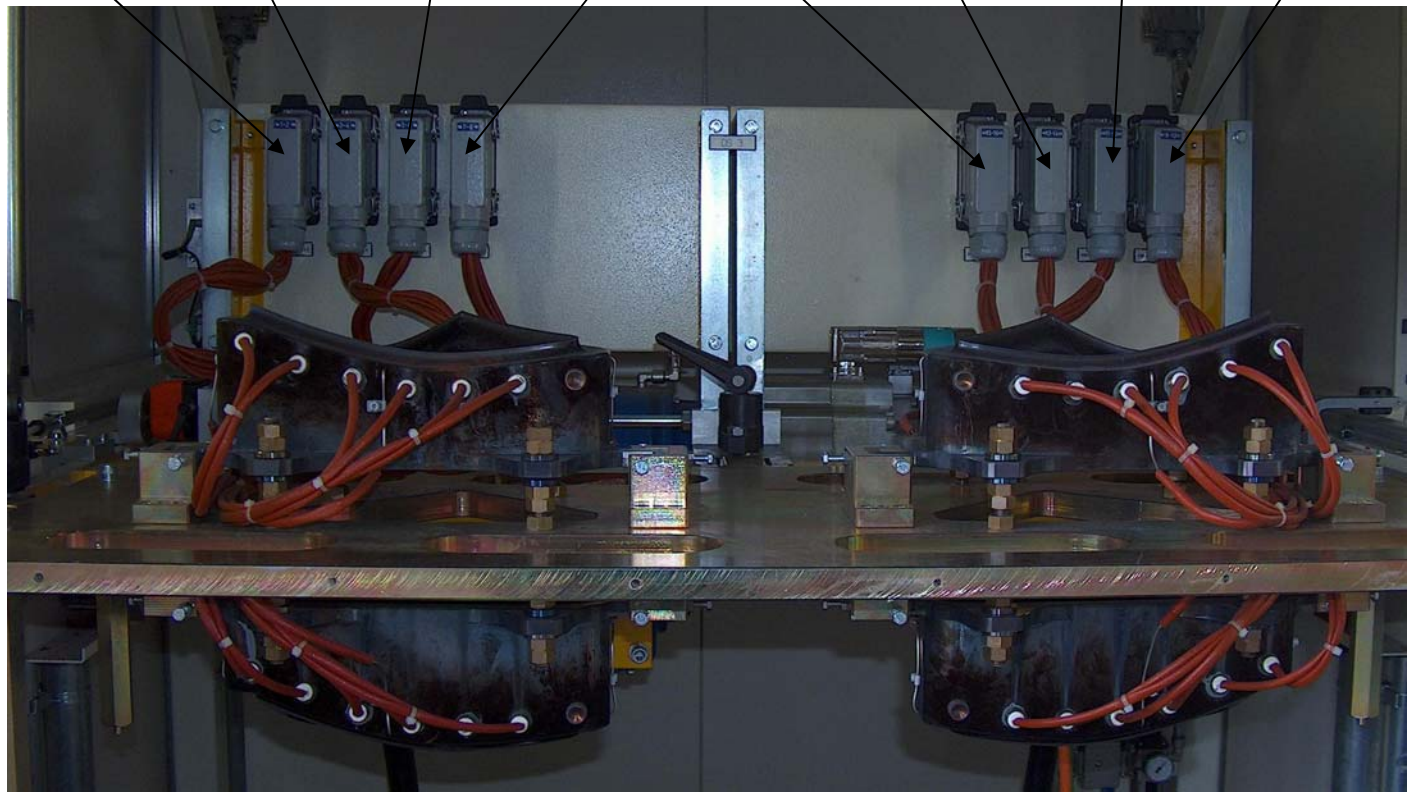


2.3.5 THERMOSETTING

The machine has been designed with nr.16 single-phase thermosetting zones.
The max. load for each zone is of about 25Amp.
Thermocouples type J.

The 16 single-phase thermosetting zones type F.I. STD are managed by selecting the HELLA STD Version in four zones from the terminal operator.

UPPER LEFT MIRROR				LOWER LEFT MIRROR				UPPER RIGHT MIRROR				LOWER RIGHT MIRROR			
Z01	Z02	Z03	Z04	Z05	Z06	Z07	Z08	Z09	Z10	Z11	Z12	Z13	Z14	Z15	Z16



2.3.6 CONNECTION FIXTURE

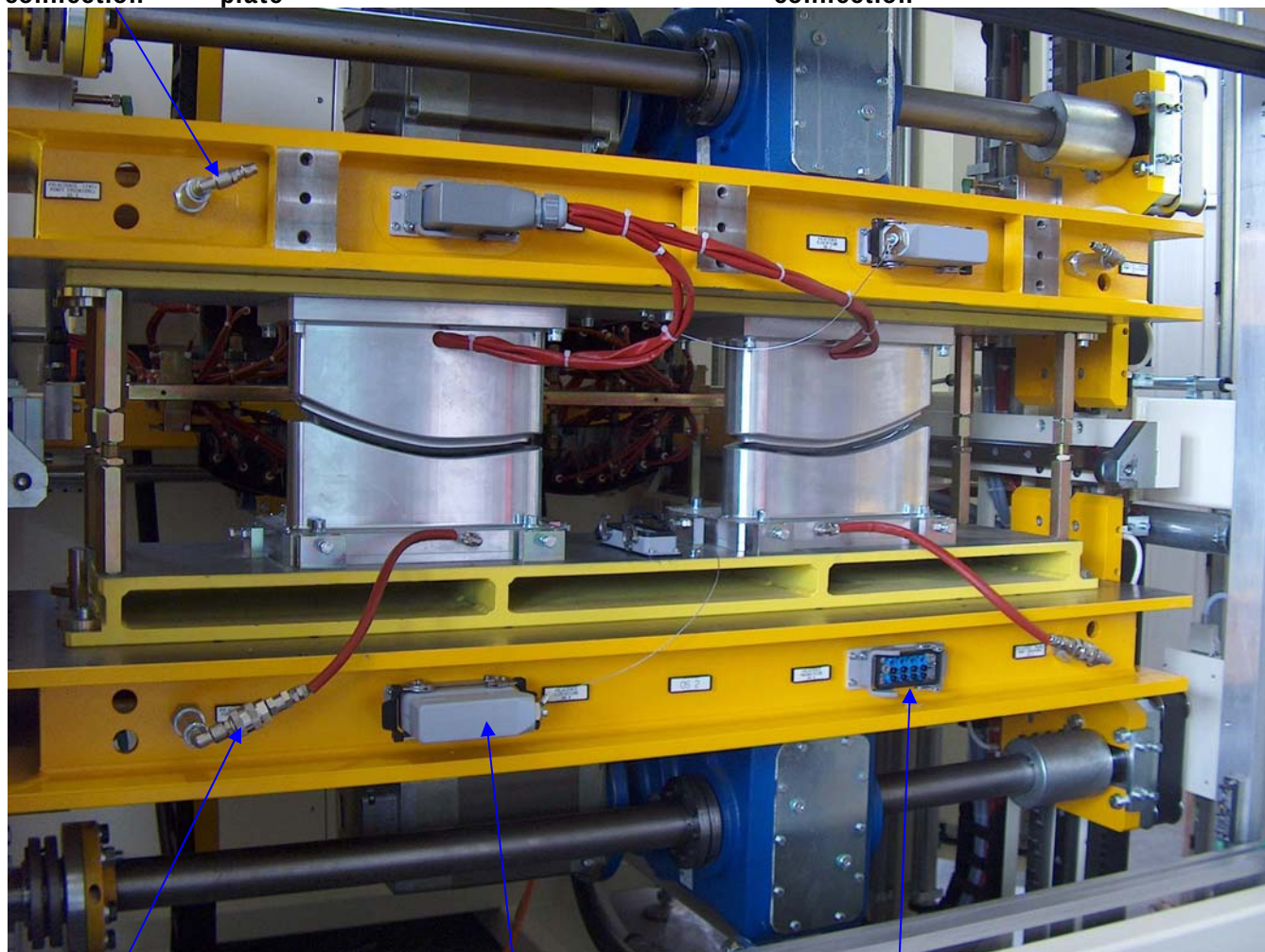
AX01 – UPPER

Left upper
vacuum
connection

Pneumatic
connection upper
plate

Electric connection
upper plate

Right upper
vacuum
connection



Left lower vacuum
connection

Lower plate electrical connection

Lower plate pneumatic connection

2.4 DESCRIPTION OF MACHINE CYCLE

The cycle of the machine is due to consecutive PHASES (PHASE) which start to the loading of elements to weld until the unload of welded components.

Each phase is programmable by terminal operator as described in the following chapters.

2.4.1 PHASE SEQUENCE

Ind.	Phase	ITALIANO	INGLESE
1	C01	C01 inserimento bloccaggi automatico	C01 automatic insertion clampings.
2	C02	C02 inserimento bloccaggi con sensori	C02 insertion of clampings through sensors
3	C03	C03 inserimento bloccaggi pedale 1	C03 clampings insertion with pedal 1
4	C04	C04 inserimento bloccaggi pedale 2	C04 clampings insertion with pedal 2
5	C05	C05 inserimento bloccaggi pedale 3	C05 clampings insertion with pedal 3
6	C06	C06 inserimento bloccaggi pedale 4	C06 clampings insertion with pedal 4
7	C07	C07 avvio ciclo	C07 cycle start
10	P01	P01 chiusura cancello operatore	P010 close of operator gate
11	P02	P02 movimento assi 1/2 per prelievo automatico	P02 axis movement 1/2 for automatic pick up.
12	P03	P03 prelievo automatico (relativo)	P03 – automatic pick up
13	P04	P04 inserimento bloccaggi per prelievo	P04 clampings insertion for pick up
14	P05	P05 movimento assi 1/2 per ingresso specchi	P05 axis movement 1/2 for mirrors entrance
15	P06	P06 movimento specchio in posizione riscaldamento	P06 mirror movement for heating
16	P07	P07 movimento assi 1/2 per avvicinamento riscaldamento	P07 axis movement 1/2 for approaching to heating
17	P08	P08 riscaldamento (relativo)	P08 heating (relative)
18	P09	P09 apertura assi 1/2 dopo riscaldamento (relativo)	P09 – axes opening 1/2 after heating (relative)
19	P10	P10 movimento assi 1/2 per uscita specchi	P10 axes movement 1/2 for mirrors exit
20	P11	P11 movimento specchio in posizione riposo	P11 mirror movement in base position
21	P12	P12 movimento assi 1/2 per avvicinamento saldatura	P12 axes movement 1/2 for approaching to welding
22	P13	P13 saldatura (relativo)	P13 welding (relative)
23	P14	P14 apertura bloccaggi a fine saldatura	P14 clampings opening after welding
24	P15	P15 apertura dopo saldatura (relativo)	P15 opening after welding (relative)
25	P16	P16 movimento assi 1/2 in posizione base	P16 axes movement 1/2 to base position
26	P17	P17 disinserimento bloccaggi a fine ciclo	P14 clampings disconnection at the end of the cycle
27	P18	P18 apertura cancello operatore	P18 opening operator gate
28	P19	P19 disinserimento bloccaggi a fine ciclo	P19 clampings disconnection at the end of the cycle
29	P20	P20 reset ciclo	P20 Reset cycle

2.5 MACHINE FUNCTIONS

The machine has been designed as standard machine for the hot mirror welding of thermoplastic components.

2.6 OPERATORS

Only one operator at a time is allowed to operate the machine.

The welding operators must be instructed on respect of :

- SAFETY REQUIREMENTS AND RISKS CONNECTED TO NON-OBSERVANCE OF INSTRUCTION AS PER CURRENT HANDBOOK.
- FUNCTIONS AND CONTROLS OF THE MACHINE.
- ALARM AND SIGNALS OF THE MACHINE.



CAUTION : A different use of the machine from those foreseen in the current handbook could cause serious damage to operators and/or the machine.

2.7 SAFETY

2.7.1 EMERGENCY

The machine is equipped with n.2 fixed emergency buttons, one on the control board and one next to left **"START CYCLE"** button. The check is made by a redundant PILZ PNOZ XV2 device, which, in case of intervention, cuts power from the drives.

To restore emergency, emergency buttons must be released and the button **RESET EMERGENCY STOP** must be pressed.

This same button is equipped with a signal lamp to inform operator that emergency has been restored.

The use of a 4-ways PILZ PNOZ XV2 device has been advised by Siemens to be used with Simodrive power systems SIEMENS SINAMICS.

2.7.2 GUARDS

The machine is equipped with 4 openings (2 on the left and 2 on the right side) for inspection, setting and maintenance operations.

Both doors may be opened with a non-fixed handle and are controlled by endswitches driven by a PILZ PNOZ X3 device which, in the case of intervention, cuts off power from the drives .

To reset the guards, close the guards and press **<RESET ALARM>**.

This push-button has a signal lamp to inform the operator that guards are been reset.

2.7.3 FRONT LOAD DOOR

A pneumatic driven load door is located in the front part of the machine to allow load and unload of components. The door movement when closing is protected by a lightbarrier SICK C4000 Micro which, if activated, cuts power off the movement electrovalve.

In this case the gate opens.

A Schmersal AES 1235 equipped with safety magnetic sensors controls the position of closed gate. By closed gate the gate closing electrovalve is powered.

This way the accidental intervention of the rib by closed gate causing opening of the same and the stop of machine cycle may be avoided. The machine in working situation is completely closed to avoid material being shot to the outside and accidental contact.

2.7.4 LIGHTBARRIER

In the front part of the machine there is a lightbarrier SICK C4000 Micro to allow load and unload of components. Dangerous movements can only made when the lightbarrier is free.

2.7.5 DANGEROUS MOVEMENTS

Dangerous movements made by the machine can only made when the gate is closed. This is valid for the automatic and the manual mode.

The part clamping and the vecuum insertion are not included.

Without all the security conditions are ok (lightbarrier free, side doors closed and emergency buttons released) the plc will not work.

2.7.6 FAILURE OF POWER AND DRIVE CIRCUITS

Failures of power circuit cause the immediate stop of machine cycle.

To reset the machine it is necessary to reset the emergency by pushing the button EMERGENCY RESET.

To start the cycle press the button RESET BASE POSITION (put the blue selector on manual cycle). When the machine is in BASE POSITION, put the selector in automatic cycle and press the button CYCLE START.

2.7.7 JOG FUNCTION - MOVEMENT WITH OPEN GATE

A function of movement of the three machine axis with open gate has been studied to allow setting operations by open gate.

The movements occur through the lightbarrier Sick C4000 Micro free.

The selection of the JOG function allows the movement of the axis chosen from the operator terminal:

- Opening of the plate safety (for AX01 and AX02)
- Opening of the concerned engine brake
- Low-speed movement of the axis

2.7.8 SAFETY DEVICES ON THE VERTICAL PLATES

The upper plate (moved by axis AX01) and lower plate (moved by axis AX02) are equipped with:

- Engine parking brake
- Pneumatic safety key with return lock.

The safety key may operate all over the machine stroke and stops the movement in case of break of the following components:

- Engine shaft
- Belt transmission system

2.7.9 OVERSTROKE OF THE AXIS

All the movement axis have overstroke in max. and min. stroke position.

Overstroke are mechanical with direct entrance in the steering.

All axis are furthermore equipped with mechanical stops.

2.7.10 MIRROR HEATING CIRCUIT

The emergency circuit disconnects the mirror heating.

In accordance with the customer the heating circuit will operate also in case of open guards.

On the heating circuit has been installed a magneto thermal protection device from 125A and a safe-life of 300mA.

2.7.11 RISK OF BURNS

During normal loading and unloading operations the mirrors are located in a backward position with respect to operator area. No accidental contact may occur.

A heat guard has been inserted, which divides the load/unload zone from the heating mirrors with open gate.

2.8 SAFETY LEVELS

2.8.1 Description of components

	EMERGENCY	GUARDS	LIGHTBARRIER	OPERATOR GATE
Device :	PNOZ XV2	PNOZ X3	SICK C4000	AES 1235
Description :	Emergency stop relay and safety gate monitor	Emergency stop relay and safety gate monitor	Light barrier	Safety monitoring module
Manufacturer :	PILZ	PILZ	SICK	Schmersal
Ref Code :	KA100	KA101	KA102	KA103
Safety categ.:	category 4	category 4	category 4	category 3
According to :	EN 954- 1	EN 954- 1	EN 954- 1	EN 954-1
Document :				
Auxiliary circuit:				
Auxiliary drive.:				

2.8.2 Description of the functions

	RESET EMERGENCY		RESET DOORS	LIGHTBARRIER SICK	CLOSED OPERATOR GATE
	--	A			
Pneumatic power					
Simodrive power supply Drive enable (9-64)					
Simodrive power supply Pulse enable (9-63)					
Simodrive power supply (NS1-NS2)					
Transformer AX01 Controller enable (65-9)					
Transformer AX02 Controller enable (65-9)					
Transformer AX03 Controller enable (65-9)					
engine brake AX01					
engine brake AX02					
engine brake AX03					
YV closing operator gate					
YV clamping and vacuum					
PLC heating drive SSR					
YV safety AX01 AX02					

Active circuit : 

2.9 FURTHER RISKS



DURING MAINTENANCE AND/OR SETTING OPERATIONS WITH OPEN GATE IT IS FORBIDDEN TO OPERATE WITH THE MACHINE SWITCHED ON.



HEATING HOTPLATES HAVE A WORK TEMPERATURE OF > 300°C DURING THEIR FUNCTION:

- **DO NOT TOUCH HOTPLATES DURING LOADING/UNLOADING OPERATIONS.**
- **BEFORE STARTING ANY MAINTENANCE WORK, HOTPLATE CHANGE OR REPLACEMENT OF HEATERS DISCONNECT HEATING (MACHINE OFF) AND WAIT FOR THE TEMPERATURE TO DROP BELOW 40°C.**



ALL MAINTENANCE WORK OF THE ELECTRIC PLANT MUST BE CARRIED OUT BY TRAINED PERSONNEL ONLY



IT IS FORBIDDEN TO DIRECT OPERATE PNEUMATICS FROM ELECTROVALVES PILOT CONTROLLERS.



ALL AXIS MOVING OPERATIONS OPERATED WITH OPEN GATE MAY BE CARRIED OUT BY TRAINED PERSONNEL ONLY.



**ALL AXIS MOVING OPERATION WITH OPEN OR CLOSED GATE DOES NOT INCLUDE INTERMEDIATE STOPS ON THE MOVEMENTS, POSITIONS ARE NOT CONTROLLED BY THE MACHINE PLC.
THE OPERATOR IS SUPPOSED TO CHECK PASSAGE VOLUME DIMENSIONS.**



DURING AXIS MOVEMENT OPERATION WITH OPEN GATE:

- **ALL OPERATORS MUST BE OUT OF OPERATING AREA**
- **ALL OPERATORS TO ENTER OPERATING AREA ONLY WITH RELEASED JOG BUTTONS**
- **ALL MACHINE ALARMS DISPLAYED ON OPERATOR TERMINALS TO BE CHECKED**



**THE MACHINE CAN LOAD STANDARD FIXTURE/HOT MIRRORS.
THE CLAMPING MOVEMENT SETTINGS WHICH CAN BE DANGEROUS FOR THE OPERATOR MUST BE INCLUDED IN THE PHASES WITH CLOSED GATE**

2.10 MISTAKES TO BE AVOIDED



AVOID TO WORK WITH HEATED HOTPLATES WITHOUT LOADING COMPONENTS, BECAUSE DURING THE HOT PHASE THE VACUUM AND RETENTION DEVICES ON FIXTURES MIGHT BE DAMAGED.

LOCATE A SHIELD BETWEEN FIXTURES AND HOTPLATES.

2.11 ELECTRIC SAFETY DEVICE

Electric safety devices refer to product, machine components and operators.

The development phase, using the supplied data, has developed the following check-up systems which grants a level of protection in conformity with laws and regulations presently in force.

Short circuit	Short circuit is limited in extent and effect thanks to the automatic interruption of circuits by means of materials and devices standing thermal, electrical, dynamic stress of such event
Overload	Avoid to use automatic switches and "gG" fuses.
Protection from over voltage	The manufacturer suggests this protection due to the complexity of the system and to sensitivity to atmospheric and other over-voltage of some of the system parts. Environmental conditions determine which system of discharger should be placed upstream.
Protection from indirect contacts	The possibility of an installation in different electrical systems, the unlikely knowledge of the manufacturer of data of the upstream network make necessary that this protection must be granted by the user already from the main supply, and it is therefore not included in the delivery. Magnetothermal differential protections which are on the proposed system have been installed to give more safety in some circuits where the application is granted and even in the event of the user not mounting a differential switch, a special level of protection on some circuits remains (circuits of sockets, computer, light and any other as mentioned in the schemes).
Protection from direct contacts	It is granted by a relevant level of IP protection, by a shell on some components which can be removed with tools, by destruction of isolation of some other components, by a switch of the «door block» kind for the access to inside parts of the electrical main board.
Voltage drop	Section of conductors has been carefully designed in size and connections have been carefully made.
Safety stop	This operation when made at the level of protection mentioned in the schemes allows a fast and safe interruption of electricity and of the mechanical movement of the parts which may hazardous for people.
Power supply	In the event of general interruption of the product by a <i>main switch</i> , the customer must grant that the upstream line has a switching system in co-ordination with the above mentioned <i>main switch</i> .
Electrical system	The product can be installed in electrical systems different from the type TT and TN, for other electrical systems consult to the manufacturer.
Electromagnetic capability	Each component corresponds to the requirements of electromagnetic compatibility, is assembled according to the instruction of the producer and grants that the final product fully complies with the relevant laws

3. INSTALLATION

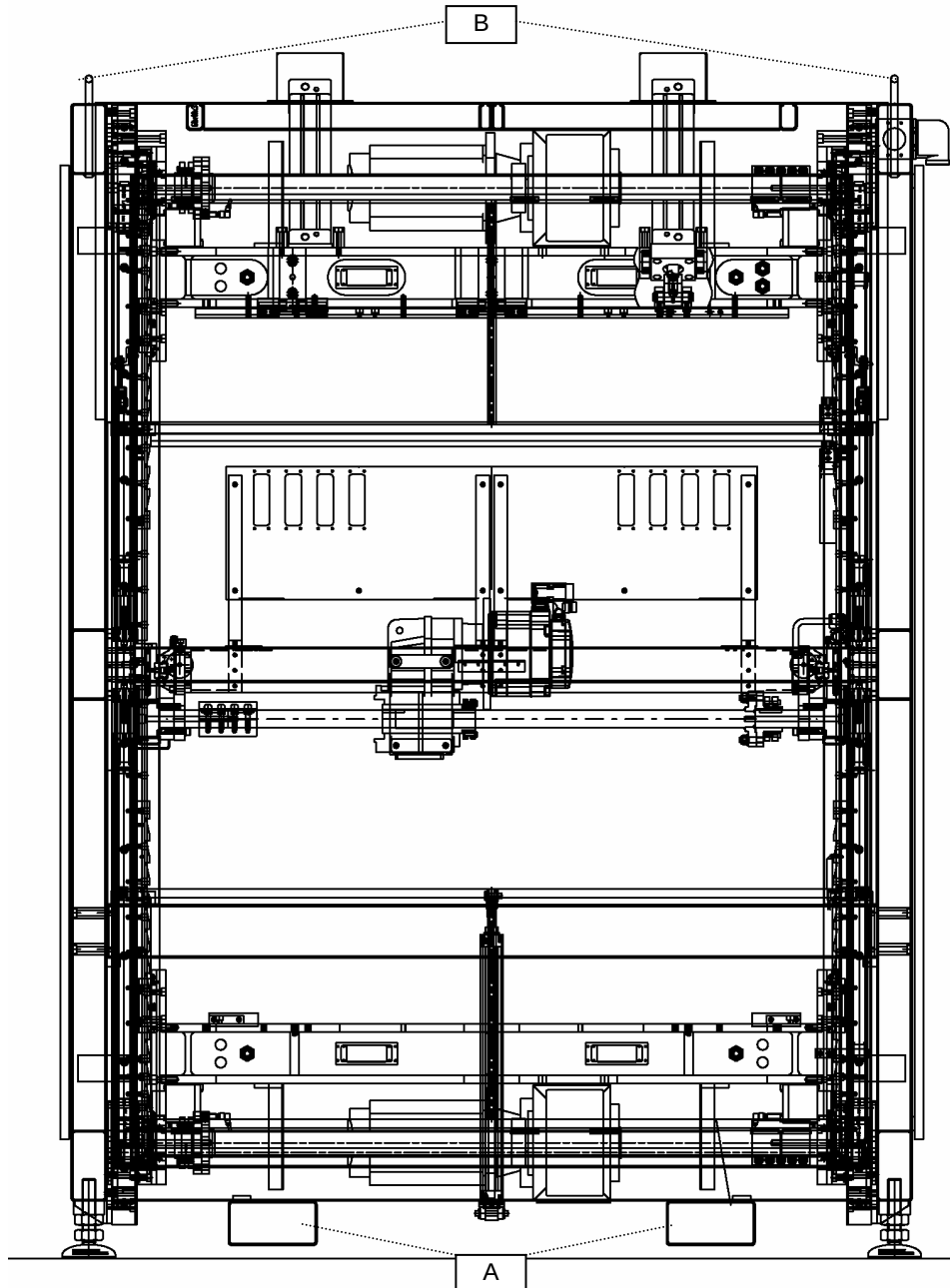
The machine has been designed for the transport without disassembly.

The machine cannot be used in USA and CANADA without authorization written from F.I.Automazione S.r.l.

3.1 TRANSPORT WITHOUT PACKING

A _ The machine is equipped of rings for the lifting and the transport with elevator

B _ The machine is equipped of rings for the lifting and the transport with crane



3.2 LONG DISTANCE TRANSPORT WITH PACKING

The machine has been packed for a long distance transport according to the way and means of transportation. Usually a wooden case is used. In this event the total mass is equal to the mass of the machine + the mass of the wooden case.

Temperature during transport must range between -20°C and +55°C to avoid damages to the machines and its components.

For higher or lower temperatures the transport system must be agreed with F.I.Automazione S.r.l.

The system of transport and the value of the mass are marked on the packing.

The lift hook must have a safety lock.

3.3 MACHINE IN WORKING CONDITION

The complete machine, placed in the final position will be in working condition only after the check-up as required by the current laws. This check-up does not establish the essential suitability of the machine, but is due to regulate its interaction with the place where the machine is placed. In fact, although the state-of-the-art making of the machine, it could be not compatible with the place of installation chosen by the customer.

Should the delivery only be part of a complex equipment system only, the final check-up asserts the suitability of the whole system and interactions among the different parts. Test has to be made taking into consideration specifications on electrical systems of machinery and specifications on low voltage electrical equipment and according to laws and decrees on electrical equipment..

When delivery includes the whole equipment, meaning the machine by this, those parts regarding check-up as per specifications on electrical equipment of machines should be excluded

When installing the machine, the Customer ought to supply the needed electrical power supply, according to the features of the machine. Co-ordination of feeding line, upstream protections, installation place and features of the machine must be taken into account.

For further technical information and specifications please contact the **MANUFACTURER**.

3.4 START UP

Place the machine according to the lay-out.

Rig the machine moving the supporting feet.

Make the following connections :

1. Electrics from the main switch IG and of ground wires
2. to pneumatic network at 6 Bar (0,6 MPa)

Check that the manometer on the filter group (area of connection to the pneumatic network) shows a pressure of 6 Bar (0,6 MPa).

Close the doors of the electrical cabinet and switch on with the main switch.

Check that all shelters are well fixed.

Check that no emergency buttons are on.

Press the RESET EMERGENCY STOP button and the corresponding green light must turn on.

3.5 POSITIONING OF THE MACHINE

To place the machine in the best position, the minimum space needed to work during maintenance and setting must be taken into account. Leave a free space on all sides of the machine, to have access to the electrical board and to shelters while making maintenance and settings. Avoid placing the machine near sources of heat since they could have a negative influx on electronic parts inside the electrical board and on control buttons.

Currents of air could negatively influence the constant temperature of mirrors.

3.5.1 MACHINE RIGGING

For a correct functioning of the machine, the correct rigging is very important. This way the machine keeps all settings made at F.I.Automazione.

Put the bubble on the plate of the machine fixtures and look for the centering, acting on the machine feet.

3.6 CONNECTIONS

3.6.1 ELECTRICAL CONNECTION

On the door of the electrical board there is a plate with indication of power supply. Please follow these indications to properly connect the feeding line of the machine.



Feeding cable of the machine must be protected from overload, short-circuit and mechanical or chemical actions which might lead to damage.

The machine must be connected to power supply by the upper terminals of the general switch **IG**.

The feeding cable must be inserted into the electrical board through a hole on the top of the board, a device to press the cable or another system must grant for a protection grade IP54 (water infiltration).

The machine must be connected to the EARTH by **PE** terminal near the GENERAL SWITCH.

3.6.2 PNEUMATIC CONNECTION

Supply to have following features:

PRESSURE 6 (0.6 MPa) BAR

To achieve the best function of pneumatic devices a dehumidifier filter is fitted.

Check the pressure gauge on filter group (connection to pneumatic web) to show a pressure of at least 6 Bar (0,6 MPa).

3.7 MACHINE DISPOSAL

Dispose of equipment and material for equipment and cleaning keeping in mind the specifications on this subjects.

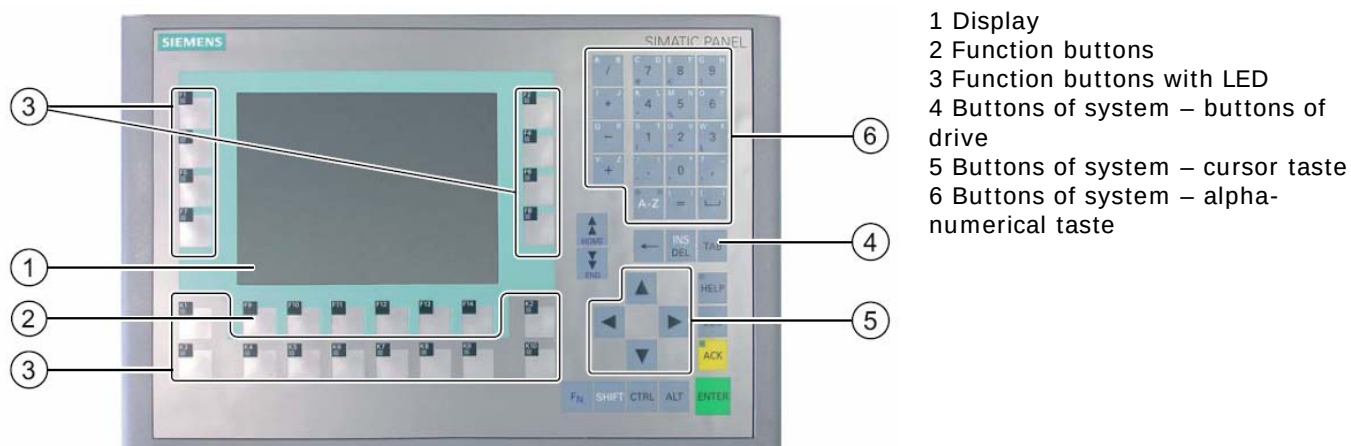
4. TERMINAL OPERATOR

The terminal operator needs to visualize and set data, to shown message about the status of the machine and the alarms.

The terminal operator is connected through PROFIBUS to the PLC of the machine; it has no data memory and all the visualized data are collected in the PLC memory.

The terminal operator is not equipped with a battery.

4.1 DESCRIPTION OF TERMINAL OPERATOR



The OP 277 is managed by the system and function buttons.

Selezione degli oggetti di comando

Tasto	Funzione	Descrizione
TAB	Tabulatore	Selezionare l'oggetto di comando precedente o successivo seguendo l'ordine delle tabulazioni.
SHIFT + TAB		
	Tasti cursore	Seleziona il primo oggetto di comando a destra, a sinistra, al di sopra o al di sotto dell'oggetto della pagina corrente. Naviga nell'oggetto di comando.

Utilizzo degli oggetti di comando

Tasto	Funzione	Descrizione
	Pagina indietro	Torna indietro di una pagina all'interno di un elenco.
F _N +	Sfoglia fino all'inizio	Sfoglia un elenco fino all'inizio.
	Pagina avanti	Avanza di una pagina all'interno di un elenco.
F _N +	Sfoglia fino alla fine	Sfoglia un elenco fino alla fine.
ENTER	Tasto INVIO	- Aziona un pulsante. - Riprende e termina l'immissione.
ESC	Annulla	- Cancella caratteri di un valore precedentemente introdotto e ripristina il valore originario. - Chiude la finestra di dialogo attiva.
INS DEL	Cancellazione di caratteri	Cancella il carattere che si trova a destra della posizione attuale del cursore.
	Cancellazione di caratteri	Cancella il carattere che si trova a sinistra della posizione attuale del cursore.
ALT +	Apri la lista di selezione	Apri una lista di selezione
CTRL + ENTER	Applica un valore	Riporta il valore selezionato nella lista di selezione senza chiudere la lista stessa.

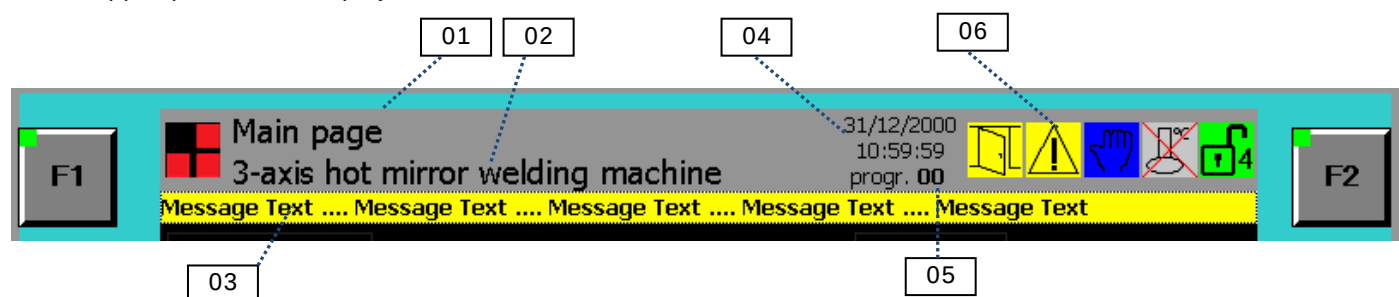
Introduzione di combinazioni di tasti

Tasto	Funzione	Utilizzo
	Commutazione della tastiera	Commuta la tastiera in caso di tasti con assegnazione multipla. - Nessun LED acceso: l'assegnazione delle cifre è attiva. Premendo una volta il tasto si ottiene la commutazione all'assegnazione dei tasti da cifre a lettere. - Un LED è acceso: l'assegnazione delle lettere a sinistra o a destra è attiva. Ogni volta che si preme un tasto si ottiene la commutazione fra l'assegnazione dei tasti a sinistra, dei tasti a destra e delle cifre.
	Commutazione maiuscole/minuscole	Utilizzo in combinazioni di tasti, p. es. per introdurre le maiuscole.
	Commutazione all'ulteriore assegnazione dei tasti	Su alcuni tasti compare una dicitura blu, p. es. il segno "%". Utilizzo delle combinazioni di tasti per l'assegnazione dei tasti di colore blu.
	Funzione di controllo generale	Utilizzo in combinazioni di tasti.
	Funzione di controllo generale	Utilizzo in combinazioni di tasti.

4.1.1 STRUCTURE OF THE PAGES

All the pages of the terminal operator are organized with a similar structure.

In the upper part of the display it is visualized:



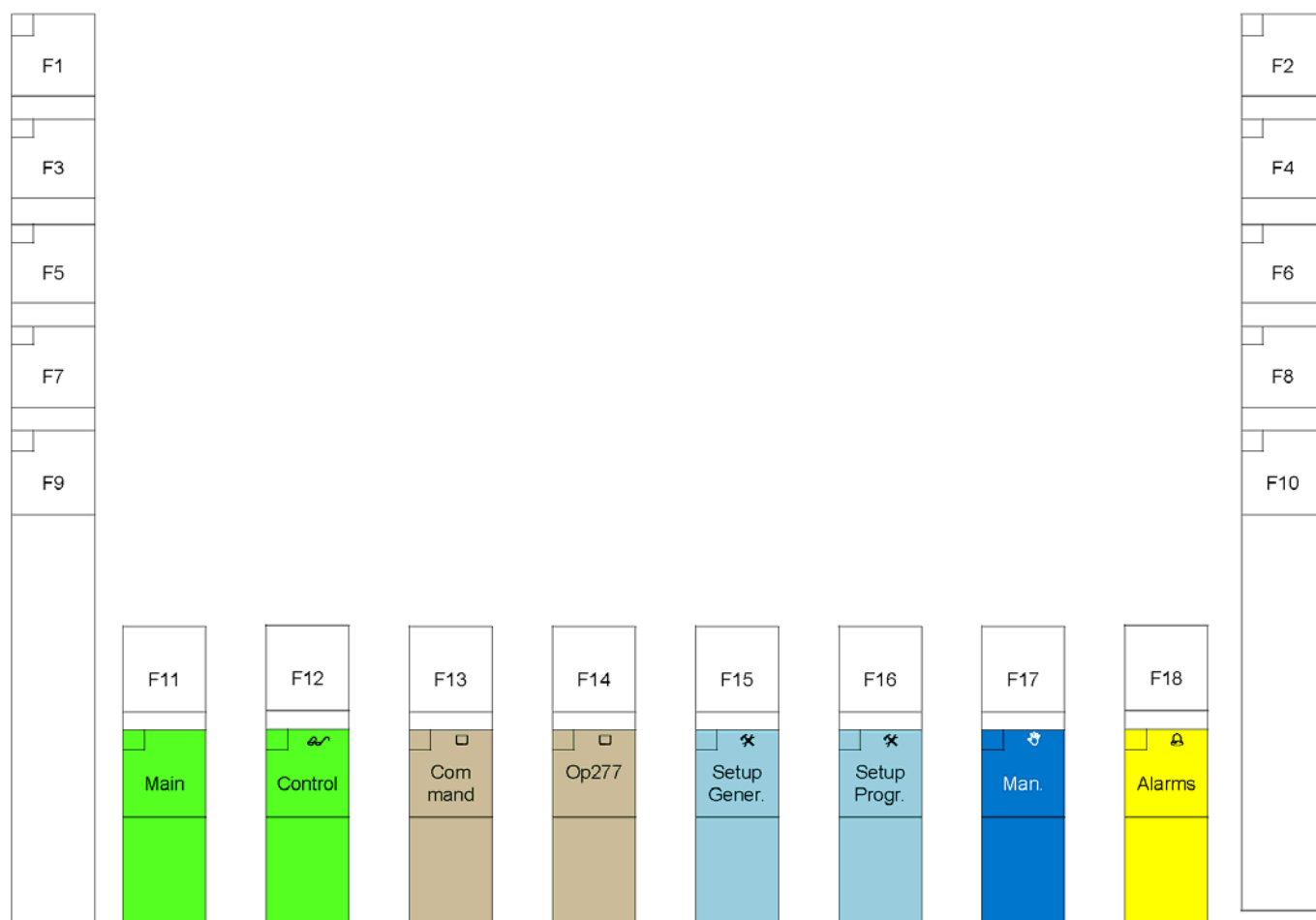
- 01 Menu title
- 02 Title of the page in the Menu
- 03 Machine alarm active
- 04 Date and hour active
- 05 Active program on the machine

06 Area graphic signaling

	Machine in emergency		Open guards or not reset
	Automatic cycle		Manual cycle
	Alarm present		Heating not lighted
	Lighted heating in alarm		Lighted and ready heating
	Password level 0		Password level 1
	Password level 2		Password level 3
	Password level 4		

4.1.2 RUNNING OF PAGES

The main page allows to enter into nr.7 Menu (indicated below).
Press the button **Home** to return in the main page.



Main
Control
Command
Op277
Setup Gener.
Setup Progr.
Man.
Alarms

Main page
 Menu for machine information
 Menu for machine commands
 Menu to set the terminal operator
 Menu for general setting out
 Menu to select/ to set out programs
 Menu to select manual commands
 Menu to visualize alarms

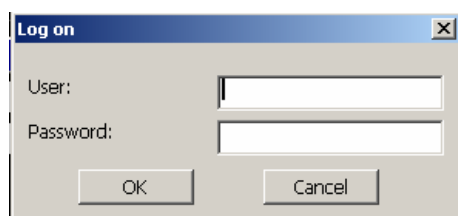
4.1.3 PASSWORD

Some pages of the terminal operator are protected by password. It is not possible to modify the value of setting, without an adequate level.

It is possible to see the actual level in all pages (icon on the top right ).

When You try to modify a data it appears a window, in which You have to insert the user name and the password. The password remains active for 5 minutes, than it vanish automatically.

To do a LOGOUT PASSWORD before the expiry of 5 minutes press the button F5 in menu Op277.



In the machine nr.5 of the following levels are active

0	Lower level, modifications are not possible
1	Level for future utility
2	Level for the modification of machine parameters
3	Level for future utility
4	Level for maintenance

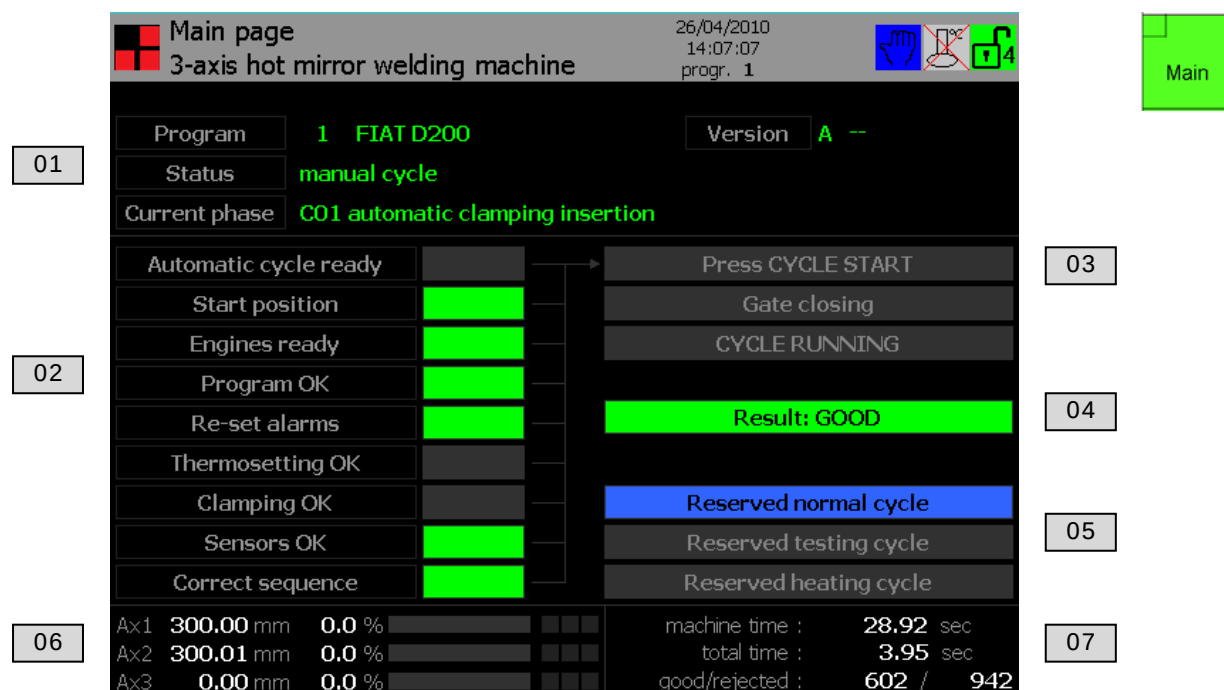
These are the following password in the machine:

Level	User	Password
0		
1	1	11111
2	2	22222
3	3	33333
4	4	44444

It is possible to change password in **Menu Op277**.

4.2 Menu MAIN – Main Page

In this page all the information about the state of the machine are shown.



- 01 Description of the active program
Description of actual program
- 02 Conditions to start the automatic cycle.
All rectangle have to be green to start the cycle
- 03 Information about the state of a machine
- 04 Result of the last welding (OK/not OK)
- 05 Type of cycle selected
- 06 Information about machine axes
- 07 Information about the production

4.3 Menu CONTROL

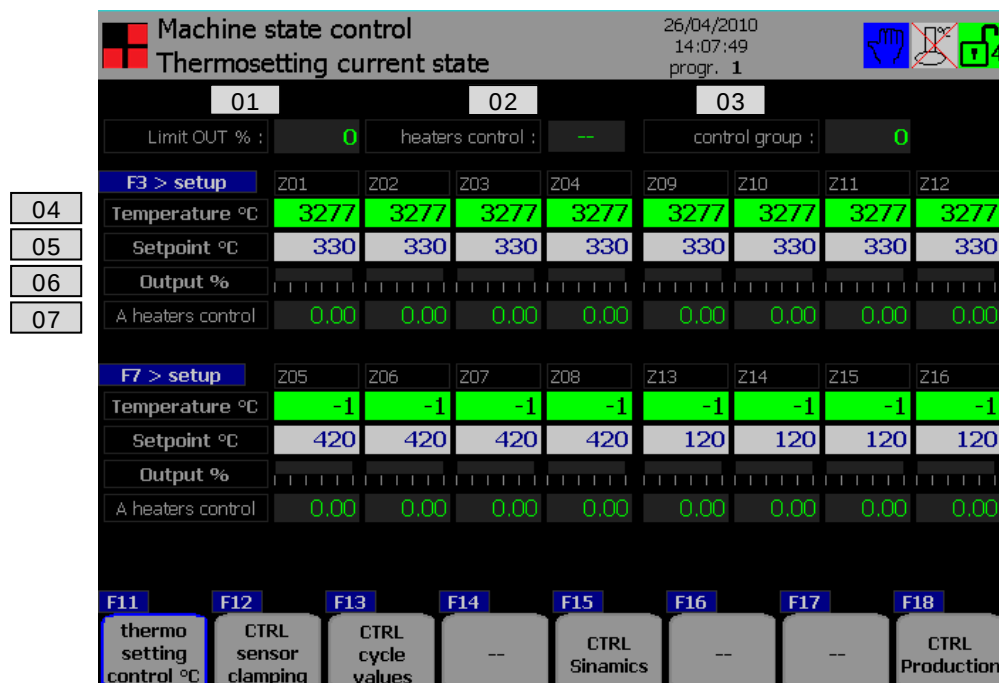
Menu CONTROL, information about machine functions

The first page of this menu is the page of thermosetting control.

To enter other pages of the menu, press the buttons F11 → F18

4.3.1 Menu CONTROL – Thermosetting page

This menu displays the actual status of thermosetting



- 01 Actual value of power limitation (pls see Softstart in the chapter Setting)
- 02 Resistance control active/not active (pls see in the chapter Setting)
- 03 Group of resistance in absorption control ((pls see in the chapter Setting)
- 04 Actual temperature
 - Green color = OK
 - Yellow color = In alarm
- 05 Set point active, in this page it is possible to modify the set point
- 06 Percentage of command of the zone (in graphic form)
- 07 Last current measure (absorption control)
 - Green color = OK
 - Yellow color = In alarm

Pressing the buttons F3 and F7 it is possible to enter directly into the pages of thermoregulation setting.

4.3.2 Menù CONTROL – Page of clampings and sensors control

Machine state control
Clamping and sensor's control

26/04/2010
14:08:15
progr. 1

01 current phase C01 automatic clamping insertion

02 drives B1 B2 B3 B4 VL VR

top axis

03 sensors S09 S10 S11 S12 S13 S14 S15 S16 L R VL VR

drives B1 B2 B3 B4 VL VR

bottom axis

sensors S01 S02 S03 S04 S05 S06 S07 S08 L R VL VR

F11 thermo setting control °C F12 CTRL sensor clamping F13 CTRL cycle values F14 -- F15 CTRL Sinamics F16 -- F17 -- F18 CTRL Production

Control

01 Indication of the actual working phase

02 Clamping

- The grey squares become green when the electrovalve of command is activated
- The indications B1→B4, VL and VR light when the programmed command is absent (see chapter Setting)

03 Sensors

- The grey squares become green when the sensor is ON
- The indications S1→S10, VL and VR light when the sensor is in a situation different from what was programmed (see chapter Setting).

4.3.3 Menu CONTROL – Page of control last cycle

Machine state control
Clamping and sensor's control

26/04/2010
14:08:36
progr. 1

Current phase C01 automatic clamping insertion

CYCLE RUNNING

Result: GOOD

cycle time : 28.92 sec

exchange time : 2.69 sec

heating time : 0.21 sec

collection force : 0.00 %

AX1 force when heating: 17.69 %

AX2 force when heating: 34.69 %

AX1 force when welding: 18.82 %

AX2 force when welding: 26.81 %

Component collection

Heating non OK

Welding force

Cycle not ended

Testing cycle performed

Heating only performed

Alarm Axis AX1

Alarm Axis AX2

Alarm Axis AX3

Alarm Sensors

F11 thermo setting control °C F12 CTRL sensor clamping F13 CTRL cycle values F14 -- F15 CTRL Sinamics F16 -- F17 -- F18 CTRL Production

Control

Visualization of last cycle results.

4.3.4 Menù CONTROL – Page of SINAMICS state control

AX1	Temperature (°C) :	14.1	01010000
Position (mm) :	299.99	Alarm code :	0
Speed (mm/min) :	0.0	Warning code :	7496
Torque 1 (%) :	0.00	Status :	00001111

AX2	Temperature (°C) :	18.2	01010000
Position (mm) :	300.01	Alarm code :	0
Speed (mm/min) :	-5.9	Warning code :	7496
Torque 1 (%) :	0.00	Status :	00001111

AX3	Temperature (°C) :	17.0	01010000
Position (mm) :	-0.01	Alarm code :	0
Speed (mm/min) :	0.0	Warning code :	7496
Torque 1 (%) :	0.00	Status :	00001111

F11	F12	F13	F14	F15	F16	F17	F18
Thermo setting CTRL °C	CTRL sensor clamping	CTRL cycle values	---	CTRL Sinamics	---	---	CTRL Production

Visualization of engine Siemens Sinamics state.

4.3.5 Menù CONTROL – Page of production control

Good components :	602		
Rejected components :	942		
Component collection :	0	Heating test cycle :	7
Heating phase :	0	Alarm Axis AX1 :	0
Exchange time :	0	Alarm Axis AX2 :	0
Welding position :	0	Alarm Axis AX3 :	0
Welding force :	0	Sensors :	0
Interrupted cycle :	4		
Testing cycle :	931		
Cycle counter :	1880	Hour counter :	457

F11	F12	F13	F14	F15	F16	F17	F18
<<	---	---	---	---	---	---	Production Reset

Counters of production

To reset the counters press the button F18.

4.4 Menu OP277

Menù OP277, allows to do all specific functions in the terminal operator.



Button F3 Password LOGIN, to visualize the window "password"

Button F5 Password LOGOUT, the operator level returns to level 0

Button F7 To visualize the page for the password modification

Button F9 It is recalled the Control Panel of terminal operator. Pls see the specific manual of Siemens Siemens OP277

Button F4 Language setting

5. MACHINE SETTING

The machine is studied to program it on the basis of the equipment.

The cycle machine depends on the settings.

The following chapters show all the pages of machine setting.

5.1 WORKING PROGRAMM SELECTION

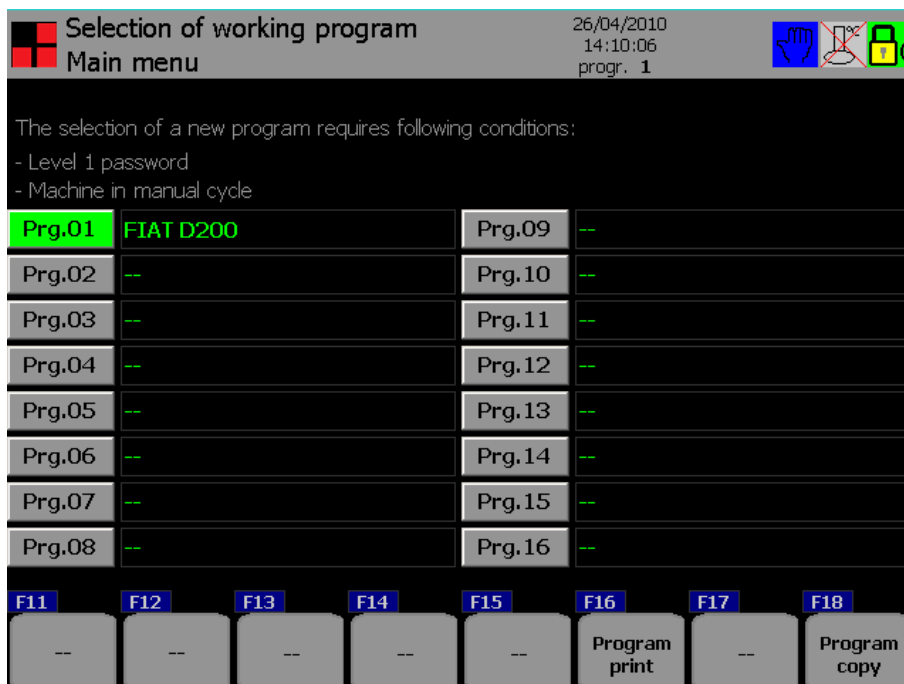
The machine can manage until 15 working programs.

In the menu **Setup Gener** it is possible to select the programs.

The active program corresponds to the green button. It is possible to see the active program in every page of terminal operator.

These are the main conditions to select a program:

- Selector CYCLE on MANUAL
- Password level greater or equal to Level 1



5.1.1 NOTE ABOUT WORKING PROGRAMMS

The data of working machine (temperature, positions, time speed, etc.) are memorized in the DATA BLOCK AREA of PLC. It is named **DB200**

When the program 1 is active, the PLC makes a copy of all data in the DB201 area

When the program 2 is active, the PLC makes a copy of all data in the DB202 area

...

When the program 15 is active, the PLC makes a copy of all data in the DB215 area

To do these copies it is not necessary to save all data.

Example: when a modification of temperature on the operator panel is done, the new temperature is memorized in DB200 and it is immediately active; moreover, the modification is memorized on the DBxxx of active program.

When You select a new program, the PLC copies from the DB of the program selected all the data in the DB200, which begin immediately actives.

Print periodically the programs (button F16) in order to have a paper copy of setting data.

Make a copy (from PLC to PG) in order to have the data of maintenance on PC.

5.2 SETTING OF FIXTURE CODE

After selecting the active program, the fixture code valid for this program must be set up.

In the page SETUP – These are the working programs:

- The program selected
- Description of program selected
- Lecture of upper program code (green color on back dark gray)
- Lecture of lower program code (green color on back dark gray)

Set up the valid fixture code for the selected program.

The fixture code are in the electrical plug (chapter 2.3.6).

There is the possibility to set different codes to manage the following working programs.

- A Solitamente si impostano i codici inferiore e superiore uguali al numero di programma
- B By setting the codes to ZERO there is the possibility to work without. This option is unwary because the machine could work without electrical plug and without equipment recognition.
- C In some cases it is possible to insert a selector on the fixture to change the code without equipment change. For example: welding selection L+R , only L, only R.

If the codes are not correct the machine gives one of the following alarms:

A014 – Program number not valid
A015 – Top fixture code not valid
A016 – Bottom fixture code not valid

If codes are not correct , the automatic cycle in the machine does not work

If codes are not correct, the manual commands of axis positioning does not work

Manual functioning of type JOG is ON.

5.3 THERMOSETTING – General

Thermosetting is managed on the three pages of terminal operator. This page regards the general setting, which are valid for all warm areas.



5.3.1 LIGHTING SOFTSTART

It is possible to start 4 type of softstart at the lighting of the inflammation. This function allows a gradual increase of command percentage to the resistances in order to avoid stress in the warm cartridge. The upper graphic shows the percentage state of Softstart.

OFF	Not activated, the command percentage starts immediately from the max. value
QUICK	The time necessary to reach the maximal value is about 2 minutes
NORMAL	The time necessary to reach the maximal value is about 8 minutes
SLOW	The time necessary to reach the maximal value is about 16 minutes

5.3.2 LIMITATION OF MAXIMAL VALUE OF COMMAND

It is possible to set up the maximal value of the percentage of command.
 By setting a value of 0% the heating is not activated.
 By setting a value of 50% the command is limited to the half of maximal power
 By setting a value of 100% the command is never limited

5.3.3 SETPOINT FOR PARKING

Setting of Setpoint (°C) for all the heating area when the selection <Heating Parking> of menu <Machine commands> is active.
 This option needs to maintain a low temperature (during the week end) for the mirrors that could be damaged because of lighting/ switching off sequence.

5.3.4 CYCLE CONTROL ABSORPTION RESISTANCES

The machine is equipped with two ammeters that point out the absorption (Ampere) of resistance, to verify if one or more resistances do not work.

The cycle control is as follow: two zone at a time are managed with regular intervals. During this phase, the absorption value is measured and memorized.

OFF
QUICK
NORMAL
SLOW

Not activated
Absorption control every minute
Absorption control every 5 minutes
Absorption control every 10 minutes

The resistance cycle control lasts 16 seconds; during this period, the PID cycle of thermosetting is not active. To have a best thermosetting it is better to set up the absorption control on SLOW.

The upper graphic shows the state of the two amperemeters measure.

5.4 THERMOSETTING – Zone Z01 → Z16

Thermosetting of each zone on two pages:

- one for upper mirrors
- one for lower mirrors



Working program setting

Thermosetting top mirror

26/04/2010

14:11:14

progr. 1





	LH top mirror				RH top mirror			
	Zone 01	Zone 02	Zone 03	Zone 04	Zone 09	Zone 10	Zone 11	Zone 12
ON/OFF	ON	ON	ON	ON	ON	ON	ON	ON
Setpoint (°C)	330	330	330	330	330	330	330	330
Alarm (°C)	15	15	32	20	15	15	15	15
Gain (%)	40	40	40	40	40	40	40	40
TI (sec)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
TD (sec)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Bias (%)	20	20	20	20	20	20	20	20
Ampere Min (A)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Ampere Max (A)	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00

F11

F12

F13

F14

F15

F16

F17

F18

<<

Thermo setting in general

Thermoset top mirror

Thermoset bottom mirror

--

--

--

--> Control Page

 Setup Progr.

Working program setting

Thermosetting bottom mirror

26/04/2010

14:11:18

progr. 1

	LH bottom mirror				RH bottom mirror			
	Zone 05	Zone 06	Zone 07	Zone 08	Zone 13	Zone 14	Zone 15	Zone 16
ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Setpoint (°C)	420	420	420	420	120	120	120	120
Alarm (°C)	15	15	15	15	15	15	15	15
Gain (%)	25	25	25	25	25	25	25	25
TI (sec)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
TD (sec)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Bias (%)	20	20	20	20	20	20	20	20
Ampere Min (A)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Ampere Max (A)	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00

F11

F12

F13

F14

F15

F16

F17

F18

<<	Thermo setting in general	Thermoset top mirror	Thermoset bottom mirror	--	--	--	--> Control Page
----	---------------------------	----------------------	-------------------------	----	----	----	------------------

ACTIVATION / DISACTIVATION

It is possible to activate or deactivate the separate heating zones

SETPOINT (°C)

Temperature as reference

ALARM (°C)

Limit of temperature +/- for alarm signaling and stop automatic cycle

GAIN (%)

PROPORTIONAL GAIN / proportional coefficient. The insertion of "proportional coefficient" shows the setting amplification.

This system eliminates the problem of temperature fluctuations using a continue setting of heating power: this is proportional to the difference between the real temperature and the set-point. For example: a big negative error produces a big tension in the heating to correct the error.

If the output power was proportional to the error of the instrument range, it would be necessary a negative error equal to half of the range to obtain the max power of the heater. Precision could be very disappointing then.

The solution is introducing the parameter of **proportional band (GAIN)**

The proportional band is usually a percentage fraction of the working interval of the device: inside the proportional band the output power is proportional to error, outside the power will be the max one or equal to zero.

Reducing the proportional band (i.e. increasing the gain), the precision of controller improves, cause even a smaller error can give a modification of output power.

But there's a limit to the increase of gain: at a certain point the system starts swinging and the gain must be reduced. A system with proportional band = zero (GAIN infinite) is a system ON-OFF

TI (sec)

RESET TIME / Time of integration. The entrance " Time of integration" determines the temporal behavior of the integrator.

In order to improve the accuracy of the proportional control the integral control is added. In a system controlled by a proportional system, with a proportional band big enough as not to cause auto- oscillation, the result is a stabile but not too accurate reading.

Assuming a residual error signal is sent to an integrator, whose exit is added to the proportional exit, the result will be an exit power increasing until the temperature reaches the set point. This way the integrator exit is cancelled and a constant power is given. The integrator could however cause oscillations. The proportional system avoids such possibility.

The integral control is feathered by the **time of integration** (integral action time), defined as the time needed by the exit to reach from zero its maximal peak in presence of a fix error corresponding to the proportional band. The RESET may be specified as a time or as a frequency (times in a minute)

In order to avoid the integral control to cause oscillations in the system, it is advisable to set the time of integration at least at the same level of the response time constant of the system.

Should the set-point be considerably varied, the integrator could be overfilled in the time the system requires to approach the new set-point, thus resulting in an overshoot once the temperature endly reaches the set point. It is therefore convenient to maintain the integrator at a zero-level until the temperature has not set down within the proportional band.

TD (sec)

DERIVATIVE TIME / Time of the shunter. The entrance "Tempo del shunter" determines the temporal behavior of the shunter

The combination of proportional and integral control guarantees a stable and precise control, all the same if the set-point is changed the system could be too slow in approximating the new set point, or alternatively quick enough but causing an overshoot.

This is controlled by the **derivative control**. As the name suggests, the derivative control measures the temporal deviation of the error signal of the system in order to reduce the speed of the variation.

The derivative control too is feathered by a typical time, the **derivative time**, (derivative action time). If the error signal is changing quickly, at the rate of a proportional band in a derivate time, then the shunter exit will be enough to take the exit power to zero.

BIAS (%)

DISTURBANCE VARIABLE / Signal of disturbance. To insert a signal of disturbance, the signal of disturbance is connected to the input " Signal of disturbance ".

Ampere MIN (A) Min value before an alarm of heaters absorption starts to sound

Ampere MAX (A) Max value before an alarm of heaters absorption starts to sound

5.5 SETTING OF AXES MOVEMENT

Menu for the setting of the machine cycle:

- Positions
- Speed
- Times

5.5.1 SETTING OF AXES - GENERAL

Page for the setting of the general parameters of axis movement.

Working program setting

Axis movement - General

26/04/2010
14:11:45
progr. 1

Override, reduction of speed in percentual	Manual cycle	Automatic cycle	Changing cycle
	10 %	100 %	2 %

Position/speed limits	Min position	Max position	Max speed
AX1 top	0.00 mm	610.00 mm	800.0 mm/s
AX2 bottom	0.00 mm	610.00 mm	800.0 mm/s
AX3 mirrors	0.00 mm	800.00 mm	1500.0 mm/s

Accelerations	Acceleration	Deceleration	Heating phase
AX1 top	50 %	50 %	100 %
AX2 bottom	50 %	50 %	100 %
AX3 mirrors	40 %	40 %	

F11

F12

F13

F14

F15

F16

F17

F18

<<

Axis in general

Automatic collection

Preparing for heating

Heating

Opening after heating

Welding

Welding end

✖
Setup
Progr.

OVERRIDE

The OVERRIDE is a general reduction of axis movement speed.

It is possible to set different Overrides according to different machine state.

Manual cycle
Automatic cycle
Cycle change

Not activated (for future using)

LIMITS OF POSITIONS AND SPEED

Setting of movement limits, these limits operate on setting of the following pages.

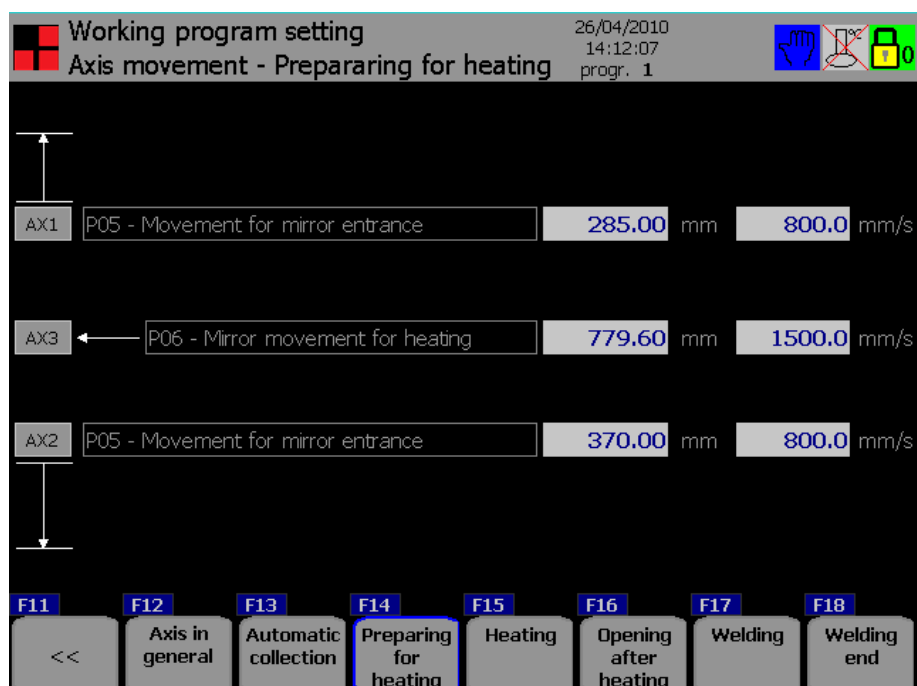
ACCELERATIONS AND DECELERATIONS

Setting of accelerations / decelerations for each axis

With a 100% setting, the axis moves with max acceleration → functioning “sudden/bad”

With a 10% setting, the axis moves with min acceleration → functioning “gentle”, machine cycle time gets longer.

5.5.2 SETTING – Axis movement – Preparing for heating



P05 P05 Axes movement 1/2 for mirror entrance

Setting of **absolute** positioning and speed for movement axes 1 (upper) and 2 (lower) before the axis 3 (mirrors) comes in.



If the setting of this position is wrong there's an high risk of collision of the axes. Let it do only to expert personnel.

P06 P06 Mirror movement for heating

Setting of **absolute** positioning and speed for movement axis 3 (mirrors) to reach heating position.

5.5.3 SETTING – Movement axes – Heating phase

Working program setting
Axis movement - heating phase

26/04/2010
14:12:24
progr. 1

AX1 ↓ Delayed movement for approaching to heating 0.000 sec
P07 - Movement for approaching to heating 439.00 mm 800.0 mm/s
AX1 ↓ P08 - Movement for heating (relative) 2.00 mm 1.0 mm/s
Heating time 4.000 sec
AX2 ↑ P08 - Movement for heating (relative) 2.00 mm 1.0 mm/s
AX2 ↑ P07 - Movement for approaching to heating 450.50 mm 800.0 mm/s
Delayed movement for approaching to heating 0.000 sec

F11 << F12 Axis in general F13 Automatic collection F14 Preparing for heating F15 Heating F16 Opening after heating F17 Welding F18 Welding end

✖
Setup
Progr.

P07 P07 Axes movement 1/2 for approaching to heating

Setting of **absolute** positioning and speed for movement axes 1 (upper) and 2 (lower) at high speed to reach the heating stage.

P08 P08 Heating (relative)

Setting of **relative** positioning and speed for movement axes 1 (upper) and 2 (lower) at low speed to reach the final heating position.

-- Heating time (sec)

The heating time starts when both axes 1 and 2 have reached the final heating position.

5.5.4 SETTING - Movement axes – Opening after heating

Working program setting
Axis movement - opening after heating

26/04/2010
 14:12:58
 progr. 1

Setup
Progr.

↑	AX1	P10 - Movement for mirror exit	285.00	mm	800.0	mm/s
↑	AX1	P09 - Opening after heating (relative)	-1.00	mm	20.0	mm/s
		P09 - Delay opening after heating (relative)	0.000	sec		
→	AX3	P11 - Resting mirror	0.00	mm	1500.0	mm/s
↓	AX2	P09 - Opening after heating (relative)	-1.00	mm	20.0	mm/s
		P09 - Delay opening after heating (relative)	0.000	sec		
↓	AX2	P10 - Movement for mirror exit	370.00	mm	800.0	mm/s

F11
<<

F12
Axis in
general

F13
Automatic
collection

F14
Preparing
for
heating

F15
Heating

F16
Opening
after
heating

F17
Welding

F18
Welding
end

P09 P09 Axes opening 1/2 after heating (relative)

Setting of **relative (negative)** positioning and speed for movement axes 1 (upper) and 2 (lower) at low speed to move away from the mirrors.

P10 P10 axes movement 1/2 for mirror exit

Setting of **absolute** positioning and speed for movement axes 1 (upper) and 2 (lower) at high speed to reach the coming out position of the mirrors



If the setting of this position is wrong there's an high risk of collision of the axes. Let it do only to expert personnel.

P11 P11 mirror movement in base position

Setting of the speed to move axis 3 (mirrors) to reach the rest position.

Position ZERO can't be modified for safety problems.

5.5.5 SETTING - Movement axes – Welding

Working program setting
Axis movement - Welding

26/04/2010
14:13:29
progr. 1

Setup
Progr.

↓

AX1

↓

AX1

↑

AX2

P12 - Movement for approaching to welding

P13 - Welding (relative)

Welding time

Min limit of welding force

Max limit of welding force

P12 - Movement for approaching to welding

589.60 mm

1.00 mm

12.000 sec

0.00 %

100.00 %

600.00 mm

800.0 mm/s

20.0 mm/s

800.0 mm/s

F11 <<

F12 Axis in general

F13 Automatic collection

F14 Preparing for heating

F15 Heating

F16 Opening after heating

F17 Welding

F18 Welding end

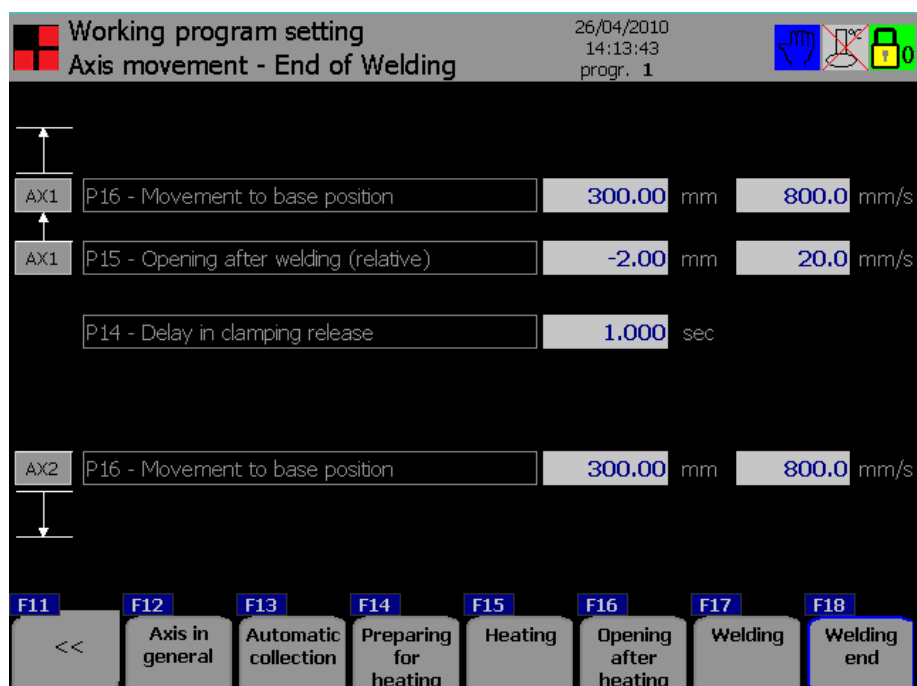
P12 P12 Axes movement 1/2 for approaching to welding
Setting of **absolute** positioning and speed for movement axes 1 (upper) and 2 (lower) at high speed to reach the welding phase (cooling)

P13 P13 Welding (relative)
Setting of **relative** positioning and speed for movement of axis 1 (upper) at low speed to reach the final welding position.

-- Welding time
The welding time starts when axis 1 has reached the final welding position.

-- MIN/MAX limit of welding force
Setting of force limits (measured on axis 2 lower) during the welding phase.

5.5.6 SETTING - Movement axes – End of welding



P14 P14 Delay in clappings release

Setting of the delay to allow clamping and vacuum release.

P15 P15 opening after welding (relative)

Setting of **relative** positioning and speed for movement of axis 1 (upper) at low speed to allow clamping and vacuum release.

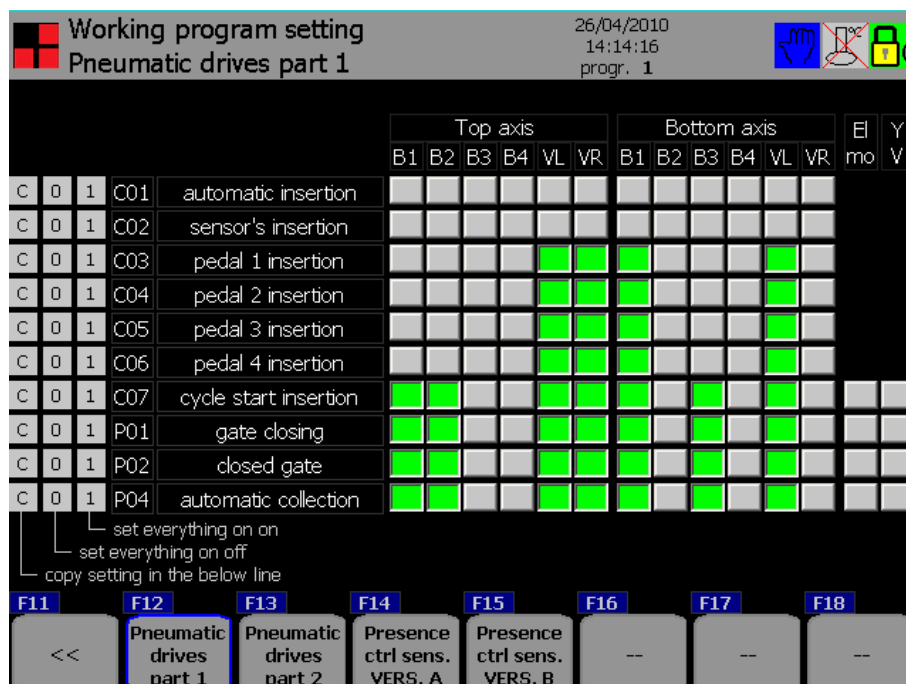
P16 P16 movement 1/2 to base position

Setting of **absolute** positioning and speed for movement axes 1 (upper) and 2 (lower) at high speed to reach the base position (cycle end)

5.6 SETTING OF CLAMPINGS

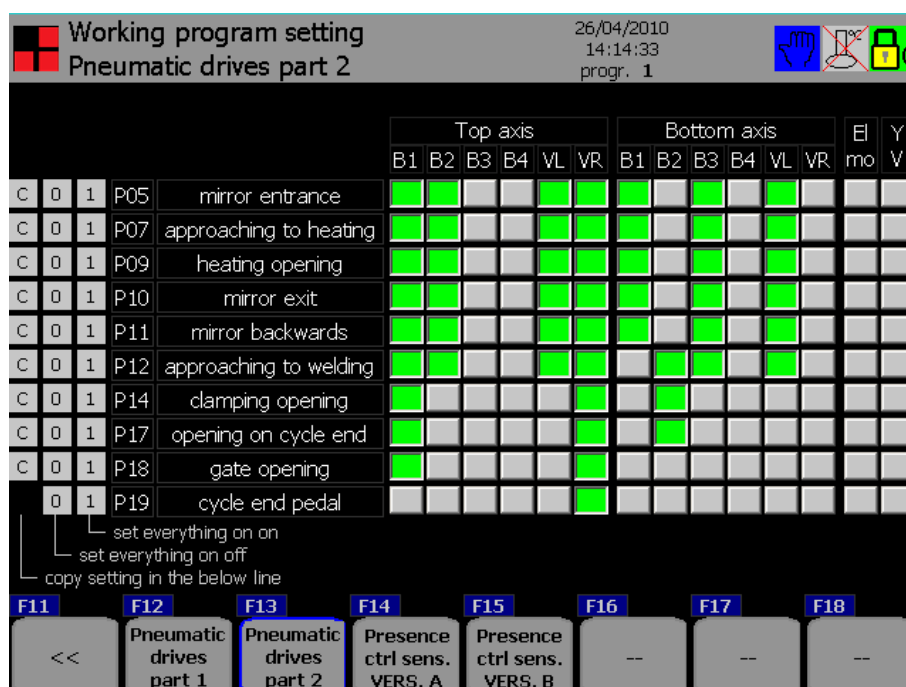
In this menu you can set clamping and sensor cycle.

5.6.1 SETTING OF CLAMPINGS – Part 1



Setting of clampings/vacuum to use in each machine phase (green color).

5.6.2 SETTING OF CLAMPINGS – Part 2



5.6.3 SETTING OF CLAMPINGS – Sensors

Impostazioni programma di lavoro 31/12/2000 10.59.59
Sensori controllo presenza progr. 00

Testo segnalazione Testo segnalazione Testo segnalazione Testo segnalazione.... Test

Asse superiore		S1	S2	S3	S4	S5	S6	S7	S8	S9	10	VL	VR
--	Sensori da verificare												
C01	inserimento autom.												
P01	per chiusura cancello												
P05	per ingresso specchio												
P20	Fine ciclo												

Asse inferiore		S1	S2	S3	S4	S5	S6	S7	S8	S9	10	VL	VR
--	Sensori da verificare												
C01	inserimento autom.												
P01	per chiusura cancello												
P05	per ingresso specchio												
P20	Fine ciclo												

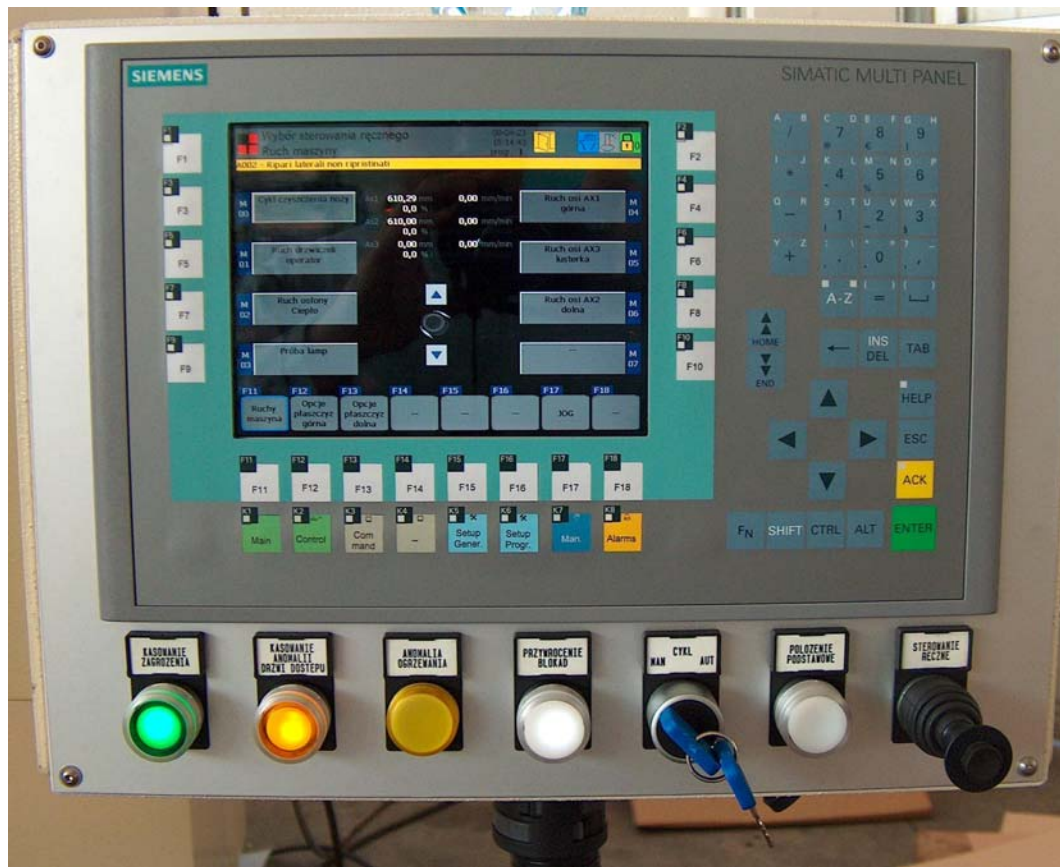
F11	F12	F13	F14	F15	F16	F17	F18
<<	Comandi pneum. parte 1	Comandi pneum. parte 2	Sensori controllo presenza	--	--	--	--

✖
Setup
Progr.

6. USE

6.1 CONTROL AND SIGNALING DEVICES

6.1.1 Control keypad



EMERGENCY RESET

Light button to reset the emergency
 Light - if OFF, the machine is OFF or in emergency
 Light - if ON, the machine is ON and the emergency is reset

RESET ALARM

Light button to restore an alarm
 Light - if OFF, the machine has no alarms
 Light - if ON, there's an alarm on the machine

THERMOSETTING ALARM

Light indicates an alarm of thermosetting

CLAMPING RESET

Light button to reset the base position of clamping/vacuum
 Light - if OFF, clampings are not in base position
 Light - if ON, clampings are in base position

CYCLE SWITCH

Switch to select the machine cycle

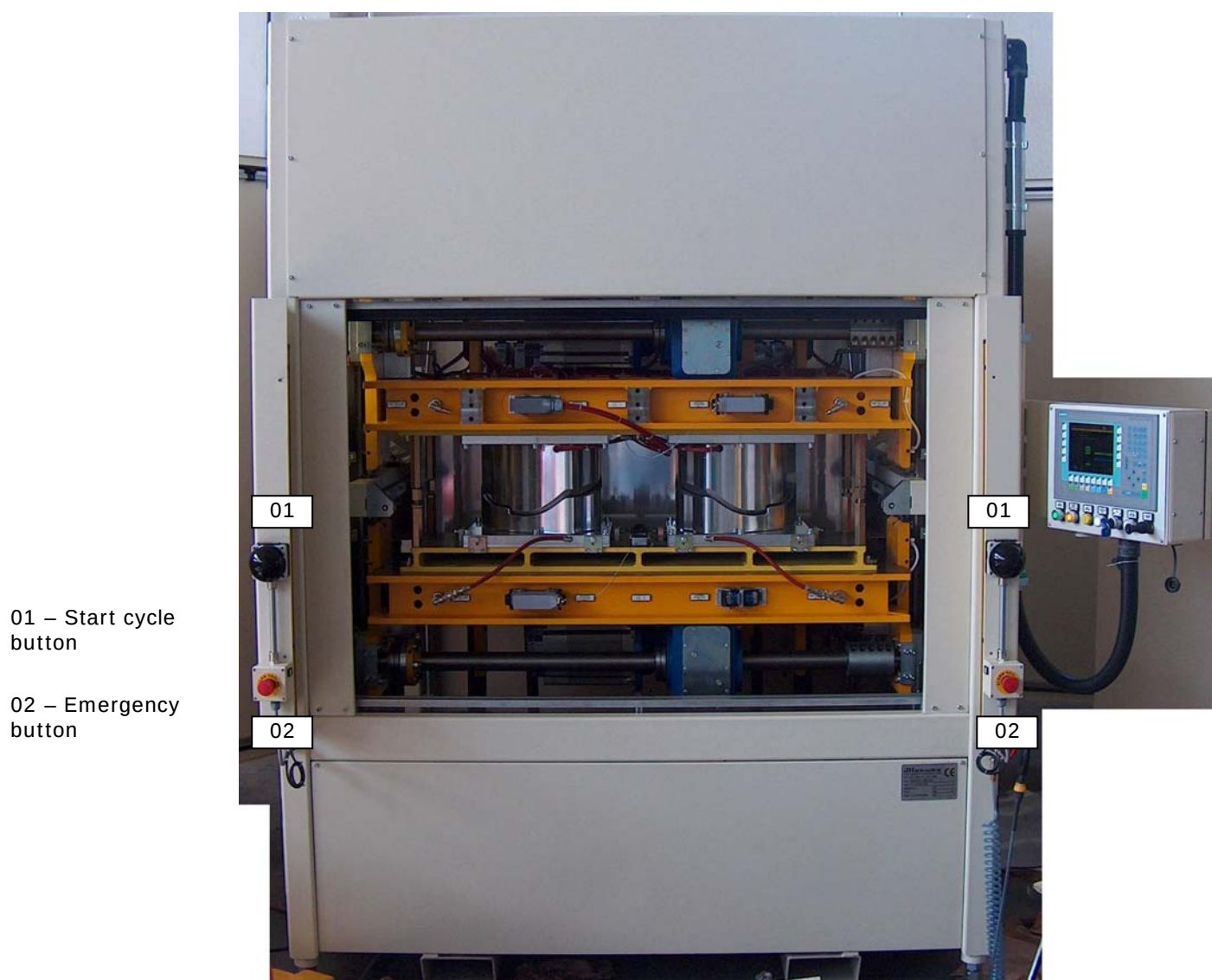
BASE POSITION

Light button to reset the base position of the machine
 Light - if OFF, the machine is not in base position
 Light - if ON, the machine is in base position

JOYSTICK

Joystick for manual controls

6.1.2 Additional signals and controls of the hardware



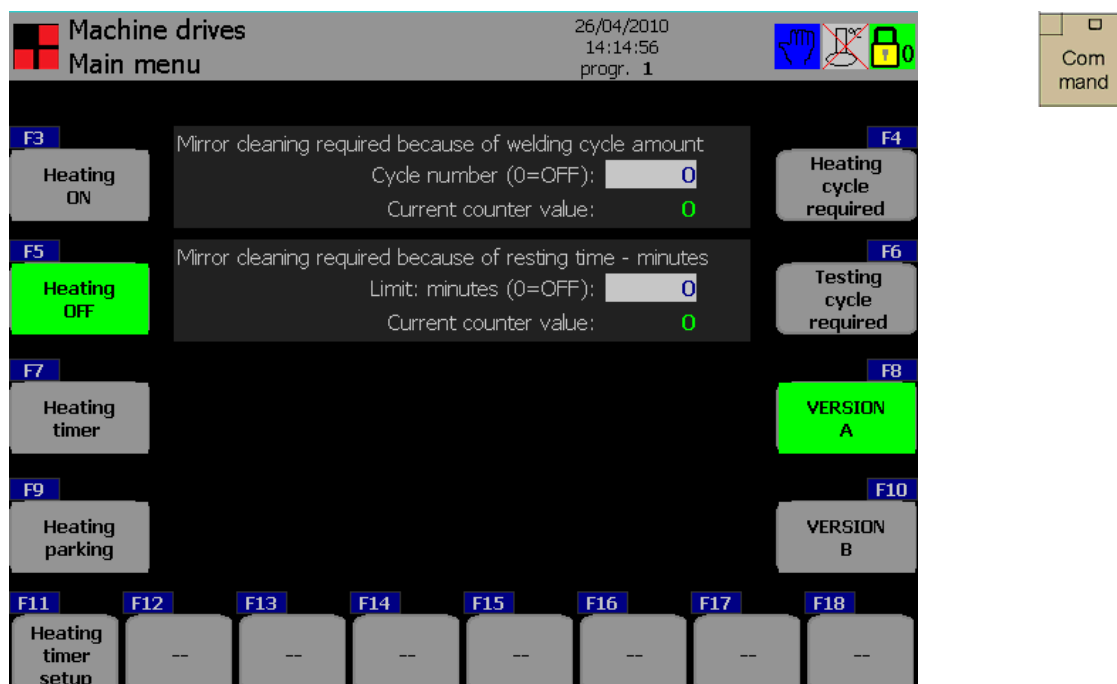
6.1.3 Signals on the terminal operator

Some signal are repeated in the top right part of the terminal operator.

	Machine in emergency		Doors opened or not restored
	Automatic cycle		Manual cycle
	Alarm		Heating off
	Heating on or in alarm		Heating on and ready

6.1.4 Controls on the terminal operator

A special menu on the terminal operator allows to select controls.



6.1.5 Controls on the terminal operator – Heating starting

Button F3	SELECTION HEATING ON
Button F5	SELECTION HEATING OFF
Button F7	SELECTION HEATING TIMER
Button F9	SELECTION HEATING IN PARKING

6.1.6 Controls on the operator terminal – Special cycles

Button F4	REQUEST OF HEATING CYCLE
Button F6	REQUEST OF TEST CYCLE

6.2 MACHINE SWITCHING ON

Check the presence of pneumatic air

Switch machine from the **MAIN SWITCH** located on the electric cabinet.

Transfer drive to main switchboard.

Press the EMERGENCY RESET key

Check the green light **EMERGENCY RESET** to be ON

Should the light be off, check :

- that the two **EMERGENCY** buttons are released.
- that the back door is closed

Press the button **RESET ALARM**

If the light **BASEPOSITION** is off,

- go to **MANUAL CYCLE**
- press the button **RESET BASE POSITION**
- Check eventual alarms on the operator terminal

Power the commander using the button **Command**

Wait for the reaching of temperature.

6.3 MANUAL CYCLE

To select to manual cycle turn the selector **CYCLE** to the position **MAN.**, wait until the corresponding blue light turns on.

In manual cycle the button **RESET BASE POSITION** is active. By pressing it the machine automatically go to base position (start position) and the lights signals **RESET BASE POSITION** will light up.

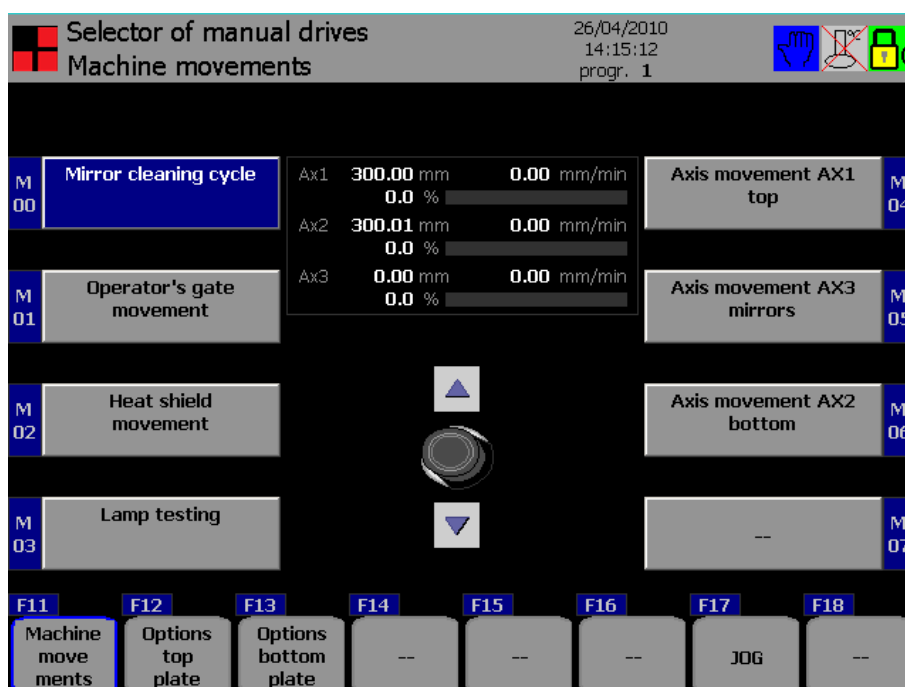
Select the command with the buttons F3 → F10

To do manual command use the JOYSTICK with four positions:

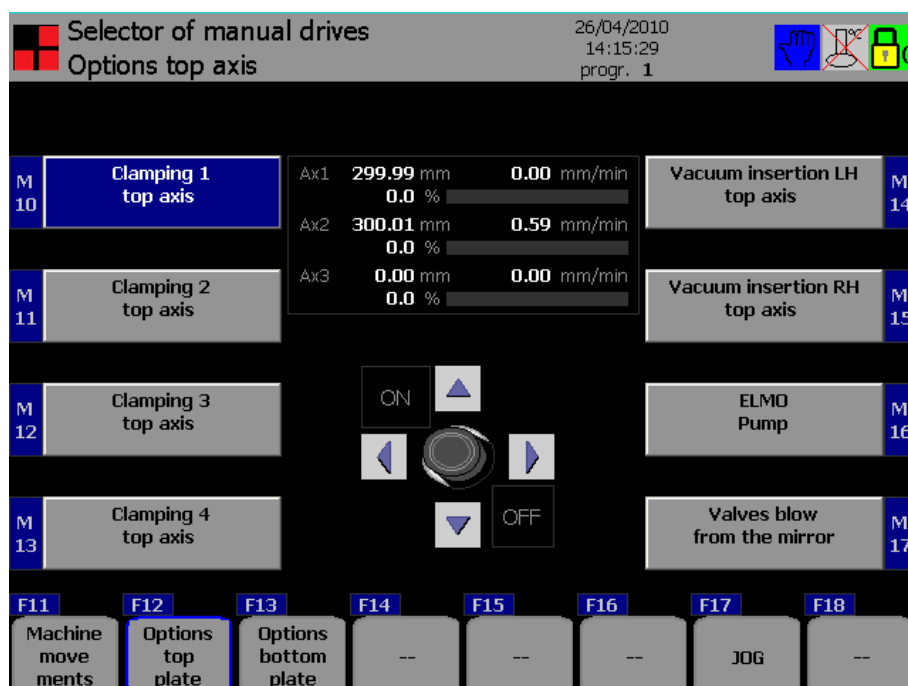
- HIGH
- LOW
- LEFT
- RIGHT



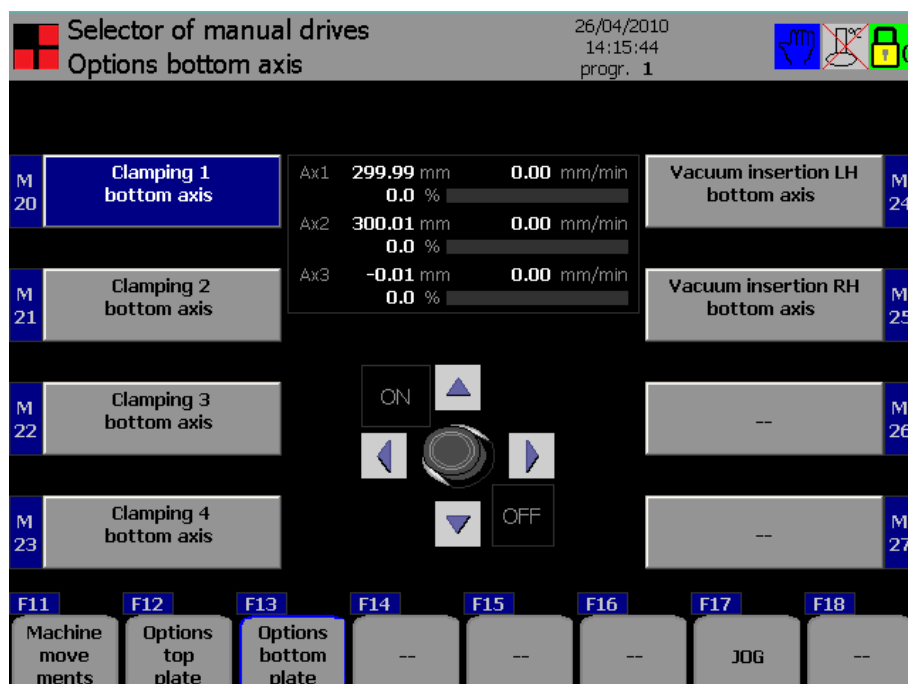
6.3.1 Manual command page 1 – Machine movements



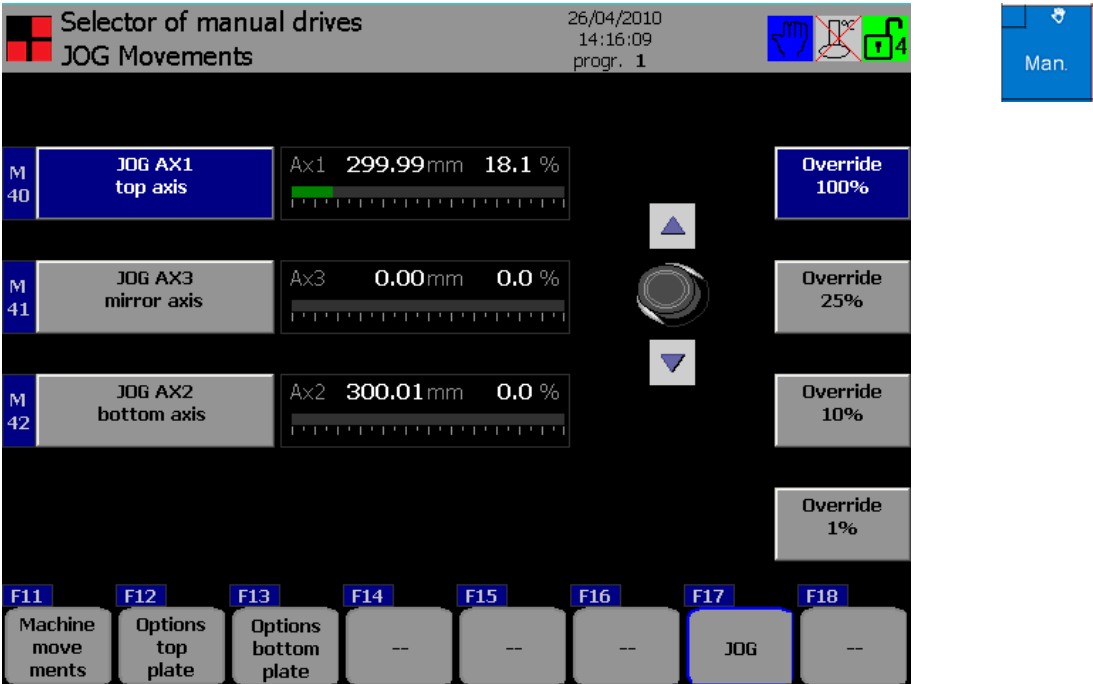
6.3.2 Manual command page 2 – Clamping and vacuum upper axis



6.3.3 Manual command page 3 - Clamping and vacuum lower axis

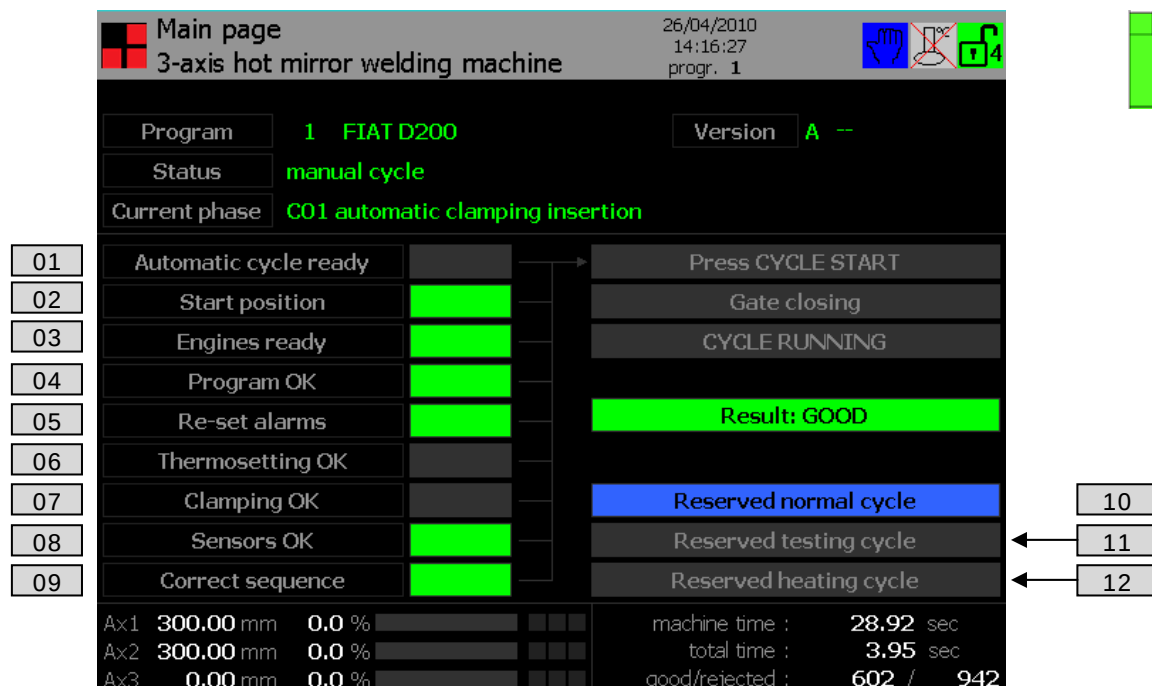


6.3.4 Manual command page 4 - JOG



6.4 AUTOMATIC CYCLE

To use the automatic cycle switch the selector **CYCLE**. You'll visualize the page MAIN.



When all the conditions for starting cycle are OK, the automatic cycle can be activated (numeration from 1 to 9). All the rectangles must be green.

- | | |
|---------------------------------|---|
| 01 Automatic cycle ready | <ul style="list-style-type: none"> - Emergency reset - Doors reset - Presence electrical power - Presence pneumatic power - Protections ready - Selector CYCLE on AUT |
| 02 Base position | <ul style="list-style-type: none"> - Machine in base position |
| 03 Engine ready | <ul style="list-style-type: none"> - No alarm Sinamics - Feeder Sinamics ready |
| 04 Program OK | <ul style="list-style-type: none"> - Fixture codes OK - Program OK |
| 05 Reset alarms | <ul style="list-style-type: none"> - No alarms |
| 06 Thermosetting OK | <ul style="list-style-type: none"> - Heating on - Heating not in parking - Temperature ok - Resistance absorption OK |
| 07 Clampings OK | <ul style="list-style-type: none"> - Clamping insertion |
| 08 Sensors OK | <ul style="list-style-type: none"> - Sensors activated |
| 09 Correct sequence | <ul style="list-style-type: none"> - Cycle ending (release of pedal elements) - The insertion phase is ended |

At this point will light the warning light **PRESS START CYCLE**

Press one of the two buttons **START CYCLE**

The operator gate closes and the automatic cycle starts.

6.4.1 AUTOMATIC CYCLE - Only HEATING

The cycle of only heating allows to verify if the “fusion” of elements to weld is correct.

The machine makes a normal cycle until the end of heating, than goes to base position and opens the gate without welding.

To start the cycle only heating :

- Open the page of terminal operator on menu **COMMAND**
- By pressing the button F4 it is possible to book the heating cycle
- In the menu **MAIN** will light the writing heating cycle booked (image in the previous page, ref.11)

The cycle heating can be reset in the following cases:

- Put the selector CYCLE in MAN position
- At the end of automatic cycle

At the end of the cycle the alarm is:

A218 - rejected component because of heating cycle

It can be reset by pressing the button RESET ALARM.

6.4.2 AUTOMATIC CYCLE – only TEST

The TEST cycle allows to make a complete welding cycle even if the heating is not ready.

To activate the TEST cycle:

- Open the page terminal operator on menu **COMMAND**
- By pressing the button F6 it is possible to book the TEST cycle
- In the menu **MAIN** will light the writing heating cycle booked (image in the previous page , ref.12)

The cycle heating can be reset in the following cases:

- Put the selector CYCLE in MAN position
- At the end of automatic cycle

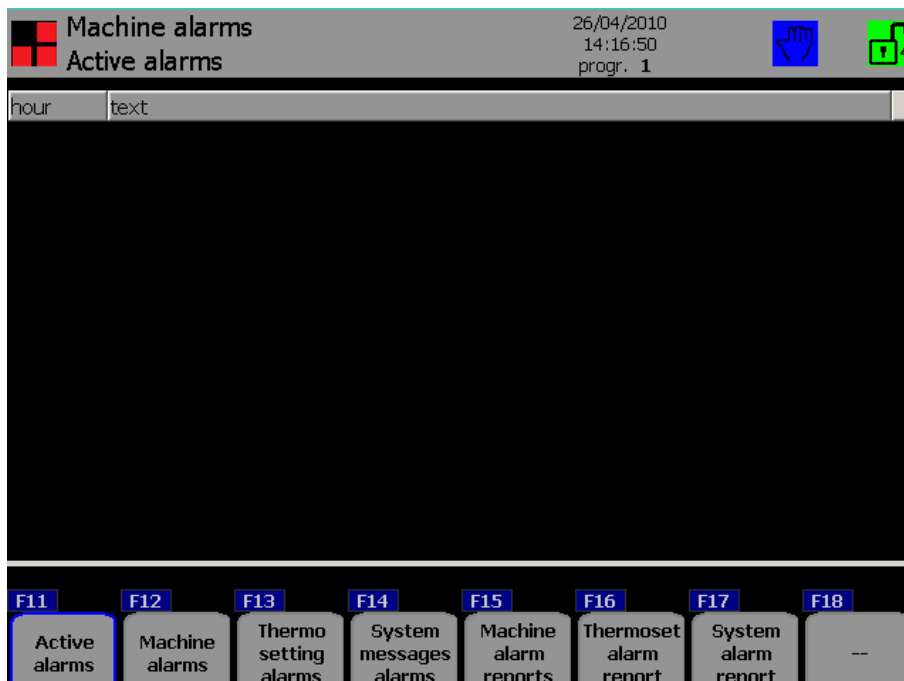
At the end of the cycle the alarm is:

A217 - rejected component because of TEST cycle. press reset button

It can be reset by pressing the button RESET ALARM.

6.5 MACHINE ALARMS

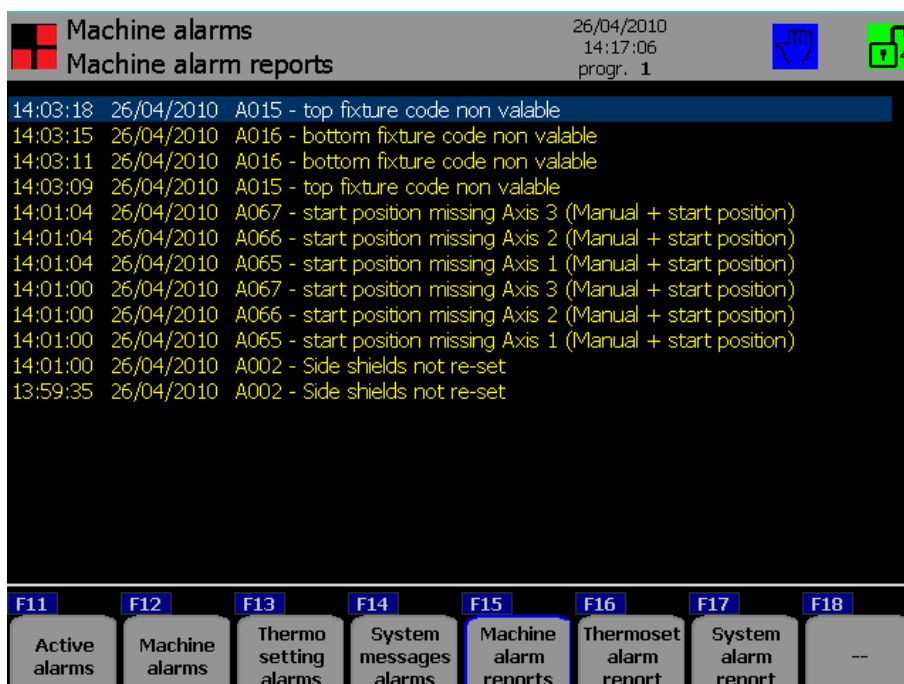
When the machine is in alarm, the light on the Control Keypad and on the terminal operator switch on.
To reset an alarm you must push the **RESET ALARM** button on the operator panel or to reset the cause.
The light **RESET ALARM** switch off only when the alarm is reset.
By pressing the button **ALARM** on the terminal operator will open the main page of alarms.



6.5.1 HISTORY ALARMS

The operator panel visualizes 3 pages about the history alarms:

- F15 → history machine alarms
- F16 → history thermosetting alarms /heating (example in the image below)
- F17 → History system alarms



6.5.2 Machine alarms – Type 1 – General

General alarms , emergency, thermic engines, etc...

A001 - emergency

Release the emergency button
Close rear doors
Press the button EMERGENCY RESET

A002 - Side shields not re-set

Close the doors
Press the button RESET ALARM

A003 - shortage of electric power

Missing electrical power in the main switch of the machine

A004 - thermal protection of heating circuit (QF9)

A005 - Thermal protection of suction engine (QM1)

A006 - Thermal protection of ELMO pump engine (QM2)

A007 - Thermal protection of Sinamics system (QM3)

A thermal protection inside the electric cabinet has tripped

A008 - Alarm: insufficient pneumatic pressure (BP)

Verify the manometer of the pneumatic plant.

A009 - alarm electric box temperature (BT)

Inside the electric cabinet you can find a thermostat which indicates a too high temperature.
Verify the thermostat setting, it must be between 40 and 45°C.
Verify that cooling fans are working and filters are clean.

A010 - malfunction of sensors : operator gate

A011 - malfunction of sensors: heat shield (SQ4.1/SQ4.2)

A012 - Misfunction of sensors: tilting head (SQ4.3/SQ4.4)

These alarms indicate that the two sensors on the cylinder/movement

- base position sensor
- work position sensor

are ON at the same time, thus it is signaled as an alarm.

6.5.3 Machine alarms – Type 2 – Programs

A014 - Program number non valable

The program selected in the menu **SETUP GENER.** is out of the admitted range 1 → 16

A015 - top fixture code non valable

A016 - bottom fixture code non valable

The codes of the plugs of both lower and upper fixtures are different from the ones set in the Menu **SETUP PROGR.**

6.5.4 Machine alarms – Type 3 – Sinamics

A017 - profibus message read with Sinamics Power supply
A018 - profibus message read with Sinamics Asse 1
A019 - profibus message read with Sinamics Asse 2
A020 - profibus message read with Sinamics Asse 3
A021 - profibus message read with Sinamics CU320
A022 - profibus message write with Sinamics Power supply
A023 - profibus message write with Sinamics Asse 1
A024 - profibus message write with Sinamics Asse 2
A025 - profibus message write with Sinamics Asse 3
A026 - profibus message write with Sinamics CU320

Alarms of Profibus communication should not be on the machine.

- Press RESET ALARM button
- Switch OFF the machine for a couple of minutes and switch it ON again
- Verify Profibus connectors, included the red terminators:
 - a) If in the Profibus connector there are 2 violet wires, put the terminator in OFF position
 - b) If in the Profibus connector there is 1 violet wire, put the terminator in ON position

Contact F.I.Automazione

A027 - Sinamics Power supply not ready
A028 - Sinamics powering axis 1 not ready
A029 - Sinamics powering axis 2 not ready
A030 - Sinamics powering axis 3 not ready

One component of the Sinamics system is not ready, press RESET ALARM button.

A033 - overrun sensor HW Asse 1 working
A034 - overrun sensor HW Asse 1 resting
A035 - overrun sensor HW Asse 2 working
A036 - overrun sensor HW Asse 2 resting
A037 - overrun sensor HW Asse 3 working
A038 - overrun sensor HW Asse 3 resting

A sensor of endstroke is ON.

Move the signaled axis in JOG mode in the opposite way.

A041 - warning Alarm Sinamics Power Supply
A042 - warning Alarm Sinamics Asse 1
A042 - warning Warning Sinamics Power Supply
A043 - warning Warning Sinamics Asse 1
A044 - warning Alarm Sinamics Asse 2
A045 - warning Warning Sinamics Asse 2
A046 - warning Alarm Sinamics Asse 3
A047 - warning Warning Sinamics Asse 3
A049 - warning Alarm Sinamics CU320
A050 - warning Warning Sinamics CU320
A051 - warning Safety Error Sinamics CU320

Signal from one of the Sinamics components.

To see the code please go to the operator terminal page

Menu CONTROL – Page of SINAMICS state check.

The list of Sinamics alarms is in the enclosures of present handbook.

6.5.5 Machine alarms – Type 4 – No base position

A065 - start position missing Axis 1 (Manual + start position)
A066 - start position missing Axis 2 (Manual + start position)
A067 - start position missing Axis 3 (Manual + start position)
A068 - start position missing heat shiueid (Manual + start position)
A069 - start position missing Tilting head (Manual + start position)
A070 - start position missing tilting head shutter LH (Manual + start position)
A071 - start position missing tilting head shutter RH (Manual + start position)

These alarms are generated in AUTO cycle, when it's impossible to start the cycle because there's not a base position.

Pass in **MANUAL cycle**

Press **RESTORE BASE POSITION** button

Sensor of base position is broken / not set

6.5.6 Machine alarms – Type 5 – Timeout pneumatic movements

A073 - Timeout opening of safety axis 1 L (SQ5.4 must be on)
A074 - Timeout opening of safety axis 1 R (SQ5.5 must be on)
A075 - Timeout closing of safety axis 1 L (SQ5.4 must be off)
A076 - Timeout closing of safety axis 1 R (SQ5.5 must be off)
A077 - Timeout opening of safety axis 2 L (SQ5.6 must be on)
A078 - Timeout opening of safety axis 2 R (SQ5.7 must be on)
A079 - Timeout closing of safety axis 2 L (SQ5.6 must be off)
A080 - Timeout closing of safety axis 2 R (SQ5.7 must be off)
A153 - Timeout operator gate closing
A154 - Timeout operator gate opening

A movement in base or work position is requested but the position can't be reached within the max set time in the PLC.

An alarm is ON.

Sensor of base position is broken / not set

Control (electrovalve) is not working

6.5.7 Machine alarms – Type 6 – Timeout axe movement

A081 - Timeout movement axis 1 for phase P02 (automatic collection)
A082 - Timeout movement axis 1 for phase P03 (relative collection)
A083 - Timeout movement axis 1 for phase P05 (Movement for mirror entrance)
A084 - Timeout movement axis 1 for phase P07 (approaching to heating)
A085 - Timeout movement axis 1 for phase P08 (Relative heating)
A086 - Timeout movement axis 1 for phase P09 (opening after heating)
A087 - Timeout movement axis 1 for phase P10 (Movement for mirror exit)
A088 - Timeout movement axis 1 for phase P12 (Approaching to welding)
A089 - Timeout movement axis 1 for phase P13 (relative welding)
A090 - Timeout movement axis 1 for phase P14 (opening after welding)
A091 - Timeout movement axis 1 for phase P16 (Movement back to start position)
A081 - Timeout movement axis 1 for phase P02 (automatic collection)
A082 - Timeout movement axis 1 for phase P03 (relative collection)
A083 - Timeout movement axis 1 for phase P05 (Movement for mirror entrance)
A084 - Timeout movement axis 1 for phase P07 (approaching to heating)
A085 - Timeout movement axis 1 for phase P08 (Relative heating)
A086 - Timeout movement axis 1 for phase P09 (opening after heating)
A087 - Timeout movement axis 1 for phase P10 (Movement for mirror exit)
A088 - Timeout movement axis 1 for phase P12 (Approaching to welding)
A089 - Timeout movement axis 1 for phase P13 (relative welding)
A090 - Timeout movement axis 1 for phase P14 (opening after welding)
A097 - Timeout movement axis 2 for phase P02 (automatic collection)
A098 - Timeout movement axis 2 for phase P05 (Movement for mirror entrance)
A099 - Timeout movement axis 2 for phase P07 (Approaching to heating)
A100 - Timeout movement axis 2 for phase P08 (Relative heating)
A101 - Timeout movement axis 2 for phase P09 (Opening after heating)
A102 - Timeout movement axis 2 for phase P10 (Movement for mirror exit)
A103 - Timeout movement axis 2 for phase P12 (Approaching to welding)
A104 - Timeout movement axis 2 for phase P16 (movement back to start position)
A097 - Timeout movement axis 2 for phase P02 (automatic collection)
A098 - Timeout movement axis 2 for phase P05 (Movement for mirror entrance)
A099 - Timeout movement axis 2 for phase P07 (Approaching to heating)
A100 - Timeout movement axis 2 for phase P08 (Relative heating)
A101 - Timeout movement axis 2 for phase P09 (Opening after heating)
A102 - Timeout movement axis 2 for phase P10 (Movement for mirror exit)
A103 - Timeout movement axis 2 for phase P12 (Approaching to welding)
A104 - Timeout movement axis 2 for phase P16 (movement back to start position)
A109 - Timeout movement axis 3 for phase P06 (mirror entrance)
A110 - Timeout movement axis 3 for phase P11 (mirror exit)

A movement in base position is requested but the position can't be reached within the max set time in the PLC.
 The alarm will be generated.
 The programmed position is wrong.
 Speed too slow.
 Override too low.

6.5.8 Machine alarms – Type 7 –

A185 - mirror cleaning required Manual cycle
A186 - heating selector in parking position. S. page. <Command>
A187 - Heating not ready
A188 - electric power missing
A189 - Safety fire function active
A190 - Battery of 24Vdc power group in alarm

6.5.9 Machine alarms – Type 10 – Lack of a condition to start the cycle

A193 - clamping missing for cycle start
A195 - sensors missing - tip fixture for cycle start
A196 - sensors missing - bottom fixture for cycle start
A197 - sensors missing - top fixture after gate closing
A198 - sensors missing - bottom fixture after gate closing
A199 - sensors missing - top fixture for component collection
A200 - sensors missing - bottom fixture for component collection
A201 - sensors missing - top fixture for cycle end
A202 - sensors missing - bottom fixture for cycle end

6.5.10 Machine alarms – Type 11 – Welding rejects

A209 - rejected component because of alarm Axis AX1. press reset button
A210 - rejected component because of alarm Axis AX2. press reset button
A211 - rejected component because of alarm Axis AX3. press reset button
A212 - rejected component because of cycle sensor. Zmacknete reset
A217 - rejected component because of TEST cycle. press reset button
A218 - rejected component because of heating cycle
A219 - rejected component because of failure in component collection
A220 - rejected component because of welding force
A221 - rejected component because of welding position
A222 - rejected component because of cycle not correctly ended

6.5.11 Thermosetting alarms – Low temperature

T001 - temperature alarm LOW in zone 01
T002 - temperature alarm LOW in zone 02
T003 - temperature alarm LOW in zone 03
T004 - temperature alarm LOW in zone 04
T005 - temperature alarm LOW in zone 05
T006 - temperature alarm LOW in zone 06
T007 - temperature alarm LOW in zone 07
T008 - temperature alarm LOW in zone 08
T009 - temperature alarm LOW in zone 09
T010 - temperature alarm LOW in zone 10
T011 - temperature alarm LOW in zone 11
T012 - temperature alarm LOW in zone 12
T013 - temperature alarm LOW in zone 13
T014 - temperature alarm LOW in zone 14
T015 - temperature alarm LOW in zone 15
T016 - temperature alarm LOW in zone 16

6.5.12 Thermosetting alarms – High temperature

T017 - temperature alarm HIGH in zone 01
T018 - temperature alarm HIGH in zone 02
T019 - temperature alarm HIGH in zone 03
T020 - temperature alarm HIGH in zone 04
T021 - temperature alarm HIGH in zone 05
T022 - temperature alarm HIGH in zone 06
T023 - temperature alarm HIGH in zone 07
T024 - temperature alarm HIGH in zone 08
T025 - temperature alarm HIGH in zone 09
T026 - temperature alarm HIGH in zone 10
T027 - temperature alarm HIGH in zone 11
T028 - temperature alarm HIGH in zone 12
T029 - temperature alarm HIGH in zone 13
T030 - temperature alarm HIGH in zone 14
T031 - temperature alarm HIGH in zone 15
T032 - temperature alarm HIGH in zone 16

6.5.13 Thermosetting alarms – Thermocouple breakdown

T033 - thermocouple alarm Zona 01
T034 - thermocouple alarm Zona 02
T035 - thermocouple alarm Zona 03
T036 - thermocouple alarm Zona 04
T037 - thermocouple alarm Zona 05
T038 - thermocouple alarm Zona 06
T039 - thermocouple alarm Zona 07
T040 - thermocouple alarm Zona 08
T041 - thermocouple alarm Zona 09
T042 - thermocouple alarm Zona 10
T043 - thermocouple alarm Zona 11
T044 - thermocouple alarm Zona 12
T045 - thermocouple alarm Zona 13
T046 - thermocouple alarm Zona 14
T047 - thermocouple alarm Zona 15
T048 - thermocouple alarm Zona 16

6.5.14 Thermosetting alarms – Heaters absorption

T049 - heaters alarm Zona 01
T050 - heaters alarm Zona 02
T051 - heaters alarm Zona 03
T052 - heaters alarm Zona 04
T053 - heaters alarm Zona 05
T054 - heaters alarm Zona 06
T055 - heaters alarm Zona 07
T056 - heaters alarm Zona 08
T057 - heaters alarm Zona 09
T058 - heaters alarm Zona 10
T059 - heaters alarm Zona 11
T060 - heaters alarm Zona 12
T061 - heaters alarm Zona 13
T062 - heaters alarm Zona 14
T063 - heaters alarm Zona 15
T064 - heaters alarm Zona 16

7. PERIODICAL MAINTENANCE

For a correct use of the machine and of its, a periodical maintenance as mentioned in the following paragraphs of this chapter is needed,

Electric as well is subject to more or less fast decay until it progressively loses the needed features of efficiency, safety, reliability and functionality.

To keep the above mentioned features at satisfying levels or to restore a good working condition of the machine, maintenance of components on the machine is needed.

Elements with the same features and of the same brand need to be used in replacement of removed components.

There are two kinds of operators who take care of the normal and special maintenance, that is the authorized worker and the skilled maintenance-man.

7.1 DAILY MAINTENANCE WORKS

The maintenance operator should:

- check the efficiency and safety of safety devices such as:
 - Emergency buttons
 - End-switches
 - self-check modules
 - lightbarriers
- replace warning lights and all the parts, when these are inefficient or damaged, with others with same features and trademark (This operation must be done with machine off).

7.2 WEEKLY MAINTENANCE WORKS

The operator should completely clean the machine.

7.3 MONTHLY MAINTENANCE WORKS

The maintenance operator should check:

- pressure values on the manometer of the pneumatic plant and, if the case, restore the right pressure (operation to be made by turned-off machine).
- check differential switches by means of the apposite trial button.
- check that no parts block the cooling of the electric engines

7.4 EVERY SIX MONTHS

The maintenance operators should

- check differential switches with test device
 - Check the electronic devices following the supplier's instructions and, in case necessary, do the necessary maintenance.

7.5 EVERY YEAR

The maintenance operator should

- check the integrity of all the machine components, with particular care to pipes and air connections, and, in case necessary, replace worn parts
- check electric plant after previously turning power off the main power
- Check:
 - cleanliness of nuts and electric clamps
 - torque of nuts and electric clamps
 - Protection conductors
 - state of the various connections
 - internal devices and general stand
- tighten nuts and clamps.

7.6 EVERY TWO YEARS

The maintenance operator should replace the accumulators incorporated in the equipment and sight-check:

- environmental compatibility of components and assembly on respect to possible mutations after installation
 - presence and integrity of data cards and protections against direct accidental damage
 - correct setting of co-ordination devices against indirect accidental damage
- wear of harness outer cover.

7.7 MAINTENANCE OF THE ELECTRICAL EQUIPMENT

CRITERIA FOR MAINTENANCE OF CONTROL BOARDS		
Main reasons of decay	Consequences	Suggested maintenance operations
Wet and dusty places	Conducting sludges lay down on clamps and their holders with subsequent superficial discharges.	Cleaning of terminal boards.
Thermal changes and vibrations .	Loosening of clamps with overheating of equipments and cable terminals.	Fasten clamps. Substitute damaged cables and equipments.
Frequent short-circuit at the limit of power interruption of automatic switches.	Possible damages on automatic switches after the 3 rd or 4 th short-circuit.	Change of automatic switches.

CRITERIA FOR MAINTENANCE OF DISTRIBUTION BOARDS		
Main reason for decay	Consequences	Suggested maintenance operations
Torn contacts after too many movements made	Heating of contacts Bad functioning	Change contacts or holder after $\frac{3}{4}$ of the normal life suggested by the maker
Wet and dusty places	Sludge on metal parts of contacts with subsequent burning of coils. Sludge on clamps and their holders with subsequent superficial discharges.	Clean metal parts. Clean clamp holders.
Short-circuits at the limit of possible values.	Possible sticking of contacts .	Substitute protecting equipment with another more limiting.
Vibrations and thermal changes	See previous board.	

PREDICTABLE LIFE OF THE MAIN ELECTRICAL COMPONENTS				
Components	Main cause of decay in normal conditions	Nominal life (a) or presumable (b)		Average presumable life expressed in years of ordinary using without maintenance
	Good use	Hours of work in normal conditions	No. of maneuvers under charge	
Wires with PVC insulation	Natural aging of the insulation	100.000 (b)		20
Wires with EPR insulation	Natural aging of the insulation	120.000 (b)		25
Maneuver switches 63 A 80 – 250 A 300 – 800 A over 800 A	Wear of contacts and mechanism		10.000 (a)	9
			3.000 (a)	5
			1.000 (a)	10
			300 (a)	10
Automatic switches for industry Up to 100 A from 100 to 315 A from 315 to 630 A from 630 to 1250 A from 1250 to 2500 A	Wear of contacts and mechanism		4.000 (a)	6
			2.000 (a)	5
			1.000 (a)	10
			500 (a)	16
			300 (a)	10
Control devices				5
Plugs (2)	Wear of plugs		5.000 (a)	4
Auxiliaries of command relays, microswitches (3)	Wear of contacts and mechanism		250.000 (b)	2
Contactors (3)	Wear of contacts and aircores		250.000 (b)	2

7.8 WARRANTY

Guarantee terms (conditions, duration, etc...) are indicated in the Contract of Sale.
Fixing in this period of time must be carried out only by the manufacturer:

F.I. AUTOMAZIONE S.r.l.
Via Sommariva 2 - 12030 Caramagna Piemonte (CN)
Tel. ++39 (0172) 89561 - Fax ++39 (0172) 89096
info@fiautomazione.com

Tampering by the user or third parties during the time of guarantee causes the end of the warranty.

8. PNEUMATIC PLANT

9. ELECTRIC PLANT

9.1.1 ELECTROMAGNETIC COMPATIBILITY

Electrical disturbance caused by electrical, electronic, electromechanical equipment mounted on board, such as :

- Programmable controllers (PLC)
- Man/machine interface systems (industrial PC)
- Control systems (diagnostic boards)
- Interfacing elements (data cards)
- Action makers (inverter)
- Electro valves
- Counters
- Engines, ecc...

might harm functionality and the duration.

To reduce this problem electromagnetic emissions in the surrounding need to be reduced in order to allow good working conditions of the above mentioned equipment.

It is necessary that each component used comply with the demands of electromagnetic compatibility, and that they are assembled according to the manufacturers instruction.

These operations grant for a final product corresponding to manufacturers specifications.

9.1.2 NUMBER OF ELECTRICAL PARTS

Electrical parts are numbered as follows :

SQ. Electrical sensors	P. Thermosetters
S. Mechanical end switches	D. Operational board
YV. Electro valve	E. Relè filters
BP. Vacuum stat	GD. Feeders
XS. Multipolar plugs	BQ. Linear transducers
BT. Thermostats	BT. Thermocouple
M. Engines	EL. Lamp on board of machine
KA. Aux. Relays	QF. Automatic switches
KM. Contactors	QM. Engine protection switch
TC. Control transformers	QS. Divider
EV. Fans	QU. Divider with fuse
HL. Lamps	SB. Button
UF. Inverter	SA. Selector
SH. Light button	

9.2 PRINCIPLES OF CABLE INSTALLATION INSIDE SWITCH BOARDS

The cable cores are made in electrolytic copper strand, flexible for fixed laying, protecting covering in PVC quality R2, with conventional colouring, numbering as in wiring diagrams, working voltage up to 700 V, max. temperature of working up to 70°C. Conductors can be identified by the insulating colouring and by the reference written at the beginning and at the end of each connection. The conductor sections are standardised as follow, co-ordinating to the several currents (i.e.: for use and calibration) the protecting devices (i.e.: automatic switches, motor protectors, fuses, etc.) with cable sections always the same for corresponding currents.

By common consent and in pursuance of the regulation in force it is used the following **conductor colouring**:

Protection/earth circuits	Masses	yellow – green
Equipotential circuits	Foreign masses	yellow – green
Power circuits in alternating current	$110V \leq V_n \leq 500V$	Black
Power circuits in direct current	$110V \leq V_n \leq 500V$	Black
Control circuits (auxiliary) in alternating current	$12V \leq V_n \leq 110V$	Red
Control circuits (auxiliary) in direct current	$12V \leq V_n \leq 110V$	Blue
Interblock circuits, external voltages	$12V \leq V_n \leq 500V$	Orange

By common consent and in pursuance of the regulation in force it is used the following **conductor section** (references of current at max calibration of protection upstream or back-up, with temperature limitations and grouping in bundle provided for this kind of switch board and not for other types of installation):

Earth protection conductors	section	Same as the conductor phase up to 16 mm ² , half conductor phase for sections over 35 mm ²
Weak current (signals) ≤ 6 A	section	0,5 ÷ 1 mm ²
Current up to 6 A	section	1,5 mm ²
Current up to 10 A	section	2,5 mm ²
Current up to 16 A	section	4 mm ²
Current up to 25 A	section	6 mm ²
Current up to 32 A	section	10 mm ²
Current up to 40 A	section	16 mm ²
Current up to 63 A	section	25 mm ²
Current up to 80 A	section	35 mm ²
Current up to 100 A	section	50 mm ²
Current up to 125 A	section	70 mm ²
Current up to 160 A	section	95 mm ²
Current up to 250 A	section	120 mm ²
Current up to 400 A	section	300 mm ²

For currents over 400 A and/or bus duct connections look up the wiring diagrams.

10. MATERIAL LIST

11. ENCLOSURES